

PRÀCTIQUES I PROJECTE DE DOCKER I APACHE HTTP SERVER

PR0375/03/01



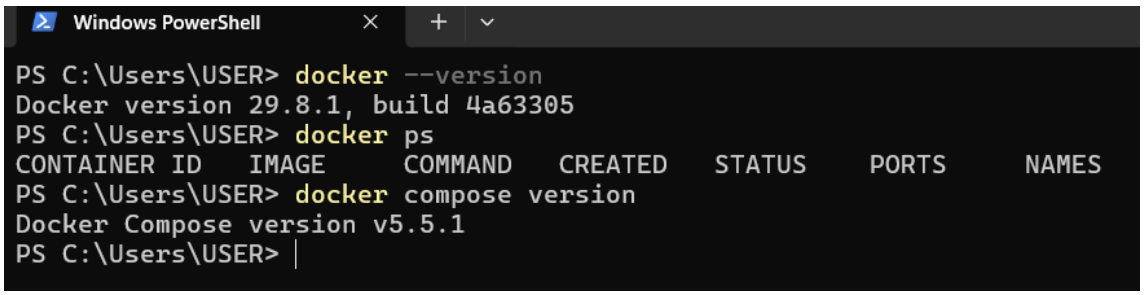
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2n ASIX

Índex

1. Verificacions prèvies 2
2. Part 1: Fonaments de Docker 2
Primer contenidor: Apache bàsic 2
Comandes bàsiques per gestionar contenidors 3
Personalitzant el contingut web 4
Inspecció de contenidors 6
Repte 1 7
3. Part 2: Dockerfile – Creant les nostres pròpies imatges 8
El nostre primer Dockerfile 8
Construir la imatge 9
Multi-stage builds (construccions en múltiples etapes) 10
Best practices per a Dockerfile 12
Repte 2 12
4. Part 3: Apache HTTP Server – Configuració bàsica 14
Mòduls d'Apache 14
Configuració personalitzada d'Apache 15
Fitxers .htaccess 18
Repte 3 20
5. Part 4: Configuració avançada d'Apache 22
Virtual Hosts - Múltiples llocs web 22
HTTPS amb SSL/TLS 27
Mod_rewrite avançat - Reescriptura d'URLs 33
Configuració de logging personalitzat 34
Repte 4 37
6. Part 5: Docker Compose - Orquestració de contenidors 41
Exemple pràctic: Apache + MySQL + phpMyAdmin 41
Executar l'stack amb Docker Compose 44
Xarxes i comunicació entre contenidors 48
Volums persistents vs Bind mounts 48
Variables d'entorn i fitxers .env 49
Repte 5 52

1. Verificacions prèvies

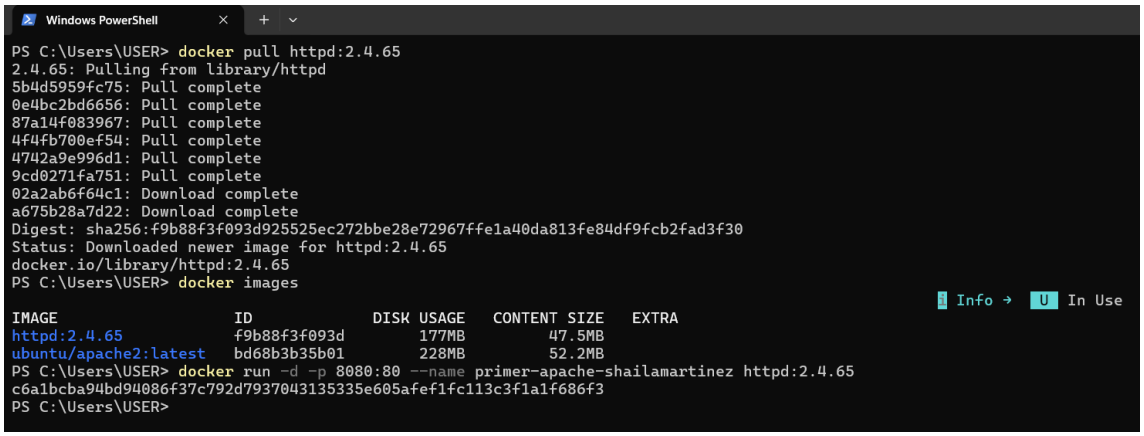
Inicialment mostro la versió del docker (29.8.1), retorna una taula buida sense contenidors i també comprovo la versió del docker compose (v5.5.1).



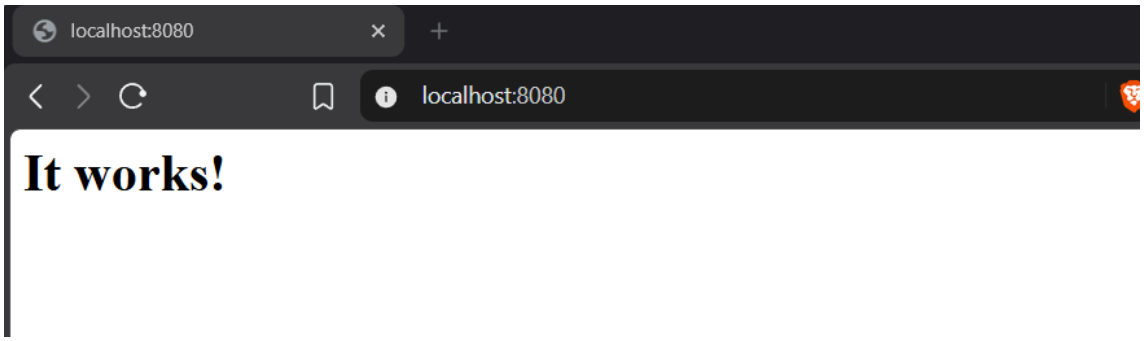
2. Part 1: Fonaments de Docker

Primer contenidor: Apache bàsic

Descarrego la imatge d'Apache, verifico la imatge descarregada i executo el contenidor.

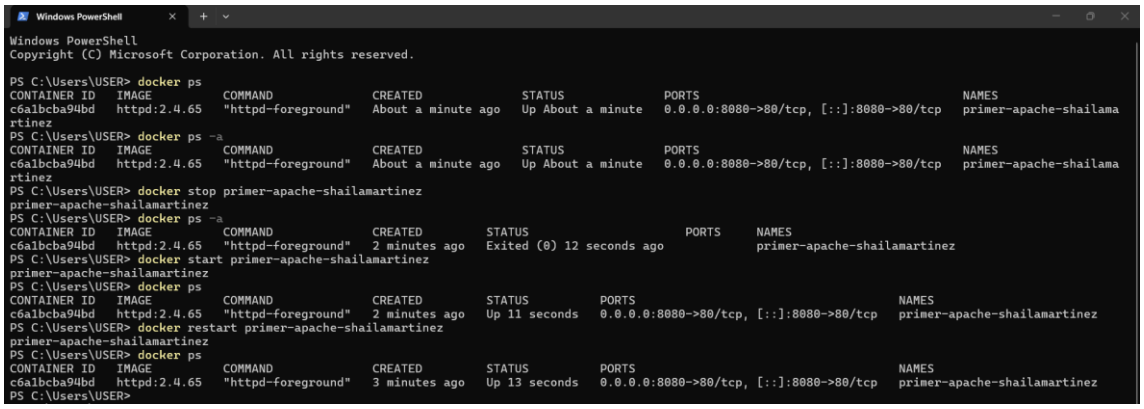


A http://localhost:8080 apareix la pàgina per defecte d'Apache It Works! Que confirma que el servidor funciona.

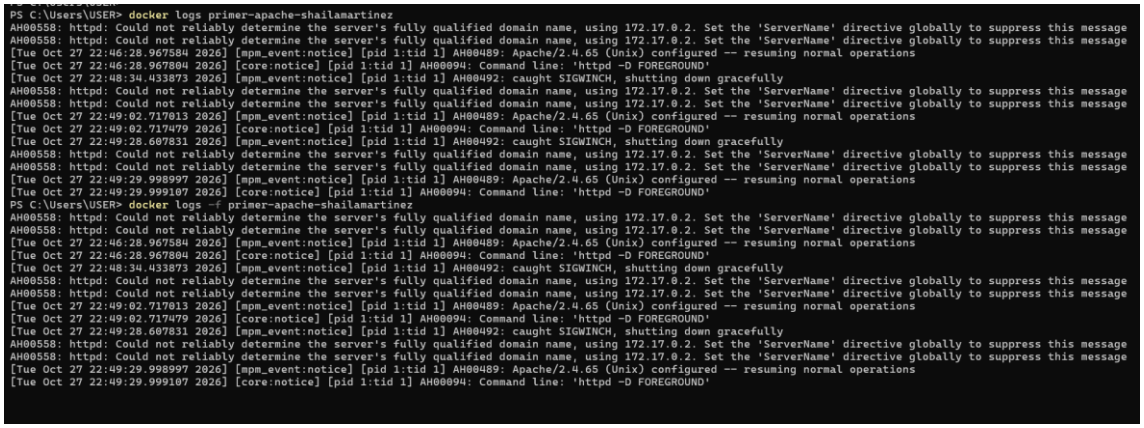


Comandes bàsiques per gestionar contenidors

Mostro els contenidors que estan en execució, si utilitzem -a es mostren tots els contenidors i els que estan aturats. Aturo el contenidor “primer-apache-shailamartinez”, després l’inicio i provo de fer docker restart.



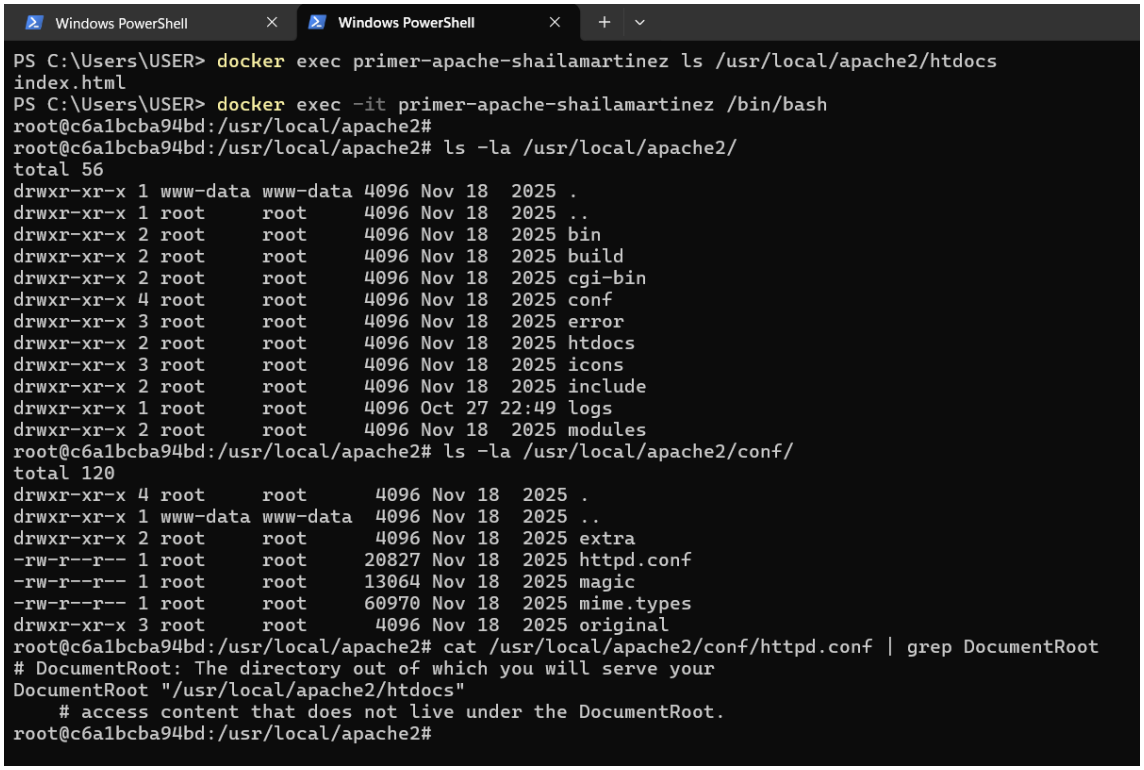
Logs del contenidor, docker logs i docker logs -f mostren els missatges d’Apache: surt un avís de que el servidor s’ha iniciat, recarregat correctament.



Exploració dins del contenidor: docker exec … ls /usr/local/apache2/htdocs mostra només index.html

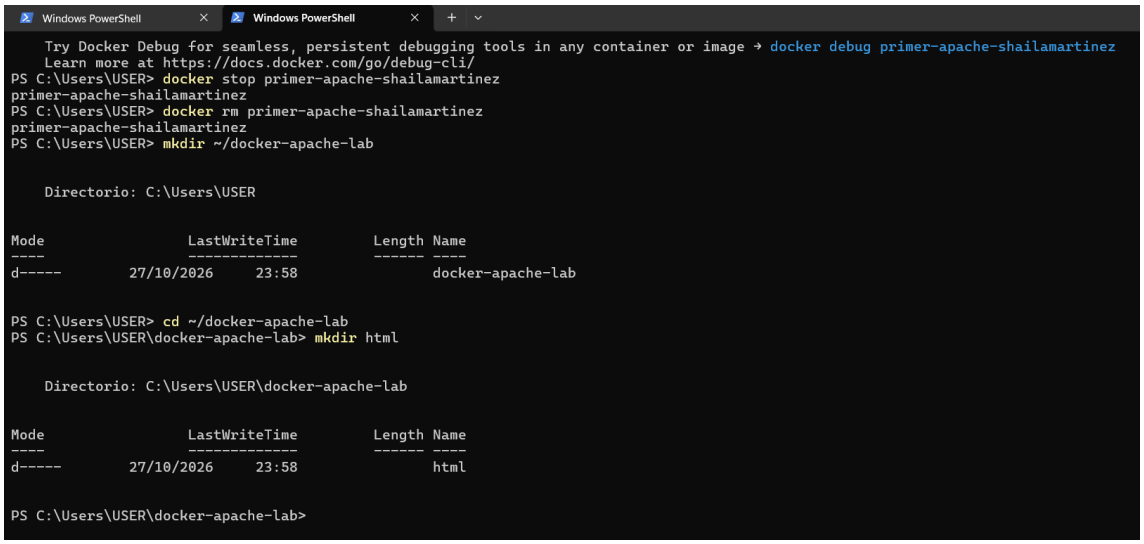
Amb -it … /bin/bash s’accedeix a una Shell interactiva dins del contenidor. Mostro el contingut de /usr/local/apache2/ i de /apache2/config

cat httpd.conf | grep DocumentRoot confirma que DocumentRoot és "/usr/local/apache2/htdocs"



Personalitzant el contingut web

Faig un stop i rm del contenidor anterior i creo les carpetes docker-apache-lab i docker-apache-lab\html (amb mkdir).



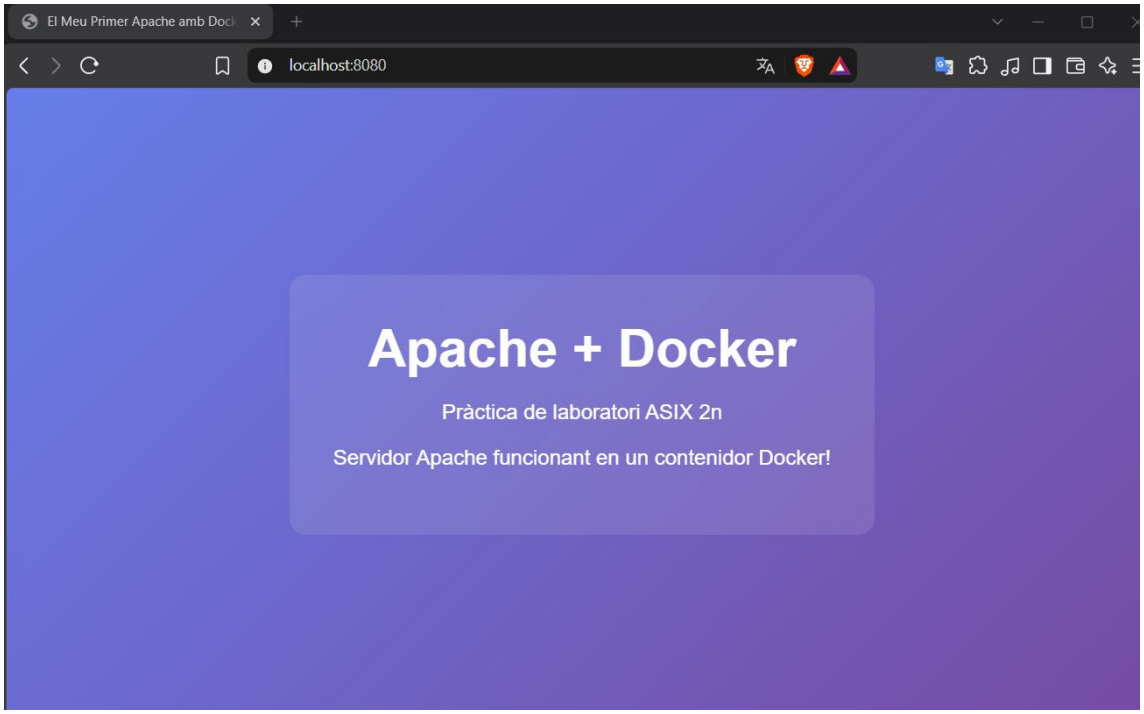
Contingut de la pàgina personalitzada (els accents no es veuen bé al powershell per la codificació).

```
PS C:\Users\USER\docker-apache-lab> cat .\html\index.html
<!DOCTYPE html>
<html lang="ca">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>El Meu Primer Apache amb Docker</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            color: white;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            margin: 0;
        }
        .container {
            text-align: center;
            background: rgba(255, 255, 255, 0.1);
            padding: 40px;
            border-radius: 15px;
            backdrop-filter: blur(10px);
        }
        h1 { font-size: 3em; margin: 0; }
        p { font-size: 1.2em; }
    </style>
</head>
<body>
    <div class="container">
        <h1> Apache + Docker</h1>
        <p>PrÃ ctica de laboratori ASIX 2n</p>
        <p>Servidor Apache funcionant en un contenidor Docker!</p>
    </div>
</body>
</html>
PS C:\Users\USER\docker-apache-lab>
```

Executo el contenidor, assignat al :8080 i mostro el htdocs on surt el html:

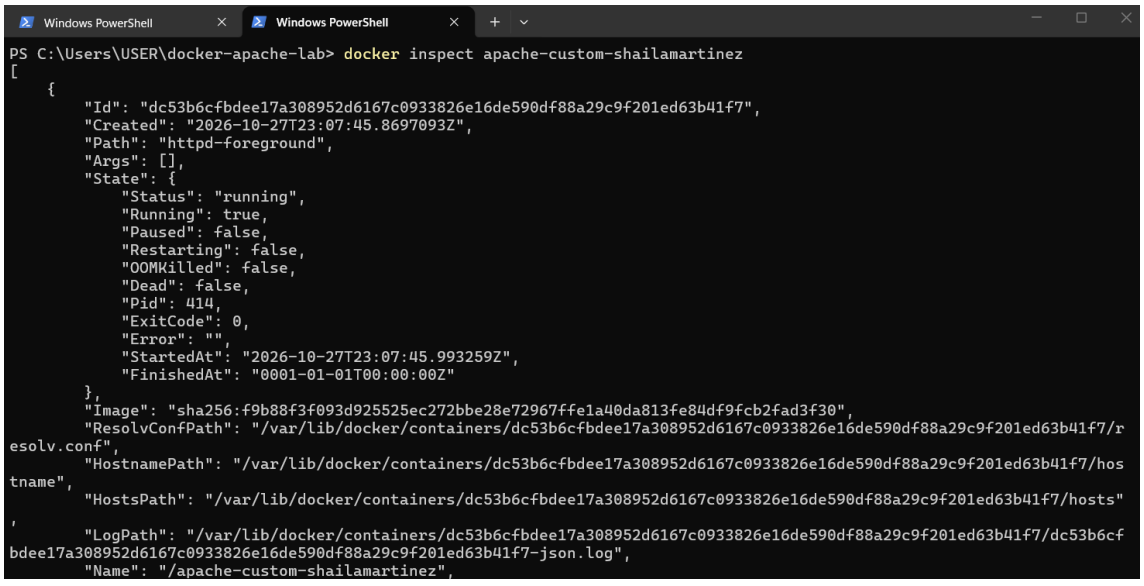
```
PS C:\Users\USER\docker-apache-lab> docker run -d -p 8080:80 --name apache-custom-shailamartinez -v "${PWD}\html:/usr/lo
cal/apache2/htdocs" httpd:2.4.65
dc53b6cfbdee17a308952d6167c0933826e16de590df88a29c9f201ed63b41f7
PS C:\Users\USER\docker-apache-lab> docker ps
CONTAINER ID   IMAGE          COMMAND              CREATED          STATUS          PORTS
      NAMES
dc53b6cfbdee   httpd:2.4.65   "httpd-foreground"   15 seconds ago   Up 14 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/
tcp   apache-custom-shailamartinez
PS C:\Users\USER\docker-apache-lab> docker exec apache-custom-shailamartinez ls -la /usr/local/apache2/htdocs
total 12
drwxrwxrwx 1 root     root     4096 Oct 27 23:03 .
drwxr-xr-x 1 www-data www-data 4096 Nov 18  2025 ..
-rwxrwxrwx 1 root     root     1075 Oct 27 23:03 index.html
PS C:\Users\USER\docker-apache-lab>
```

Al navegador provo el localhost:8080 i es pot veure que surt la pàgina personalitzada d'Apache + Docker abans creada. Qualsevol canvi es mostra a l'instant.

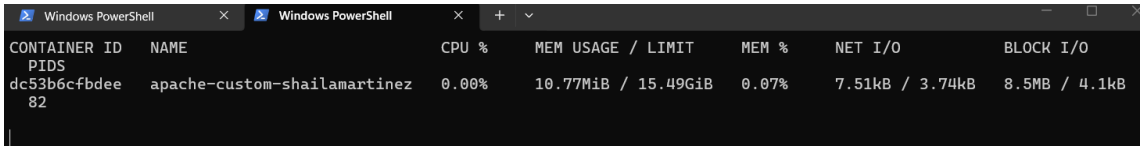


Inspecció de contenidors

Retorna el JSON complet del contenidor apache-custom-shailamartinez: status, logs...



docker stats apache-custom-shailamartinez:

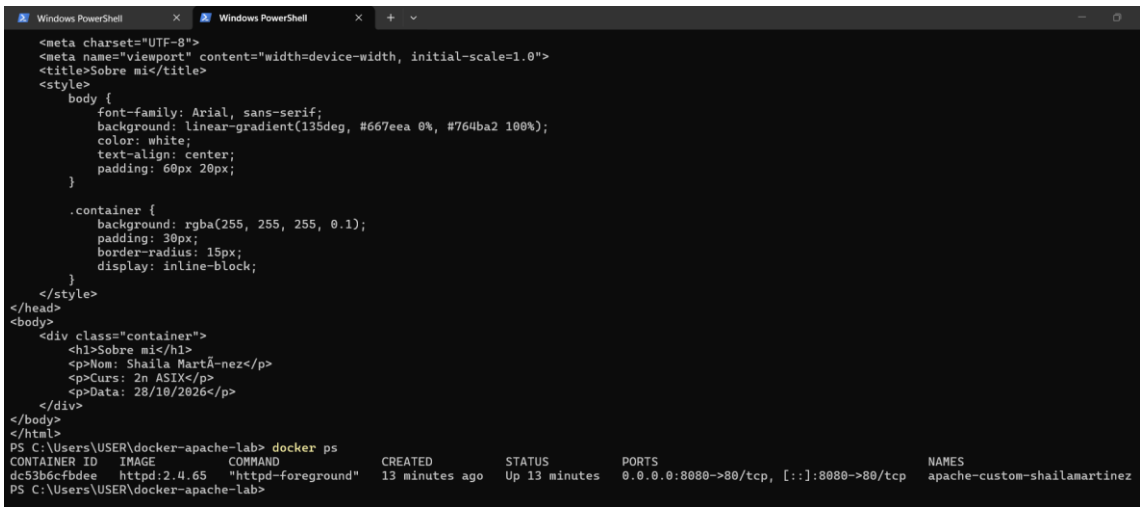


Dins del contenidor aquesta comanda permet mostrar els processos en execució que té.

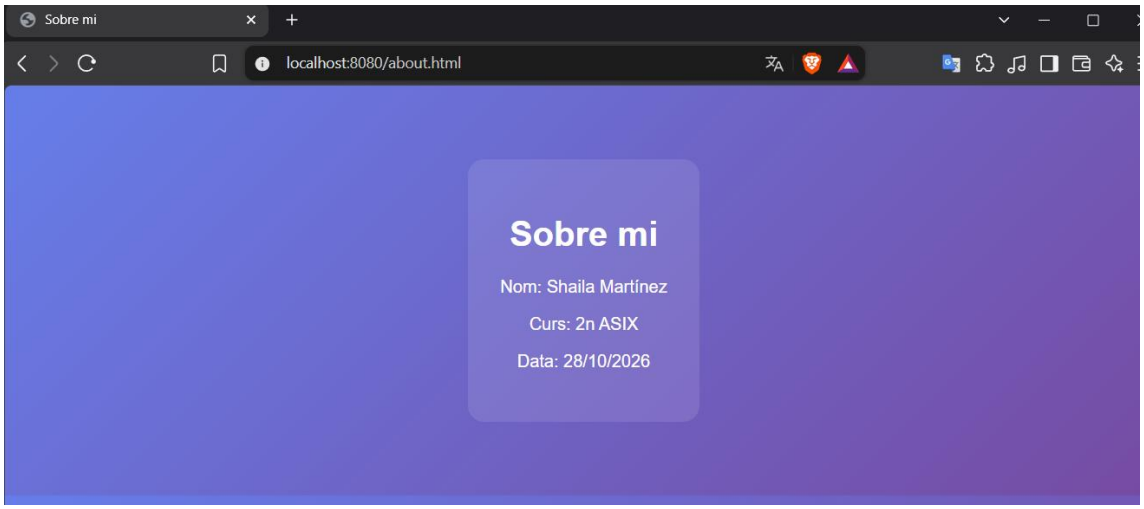
```
PS C:\Users\USER\docker-apache-lab> docker top apache-custom-shailamartinez
UID                 PID                 PPID                C                   STIME               TTY
TIME                CMD
root                414                 391                 0                   23:07               ?
00:00:00            httpd -DFOREGROUND
www-data            435                 414                 0                   23:07               ?
00:00:00            httpd -DFOREGROUND
www-data            436                 414                 0                   23:07               ?
00:00:00            httpd -DFOREGROUND
www-data            437                 414                 0                   23:07               ?
00:00:00            httpd -DFOREGROUND
PS C:\Users\USER\docker-apache-lab>
```

Repte 1

Creo el fitxer about.html a la carpeta html amb un estil. docker ps posterior mostra que el contenidor continua actiu sense haver-lo reiniciat.



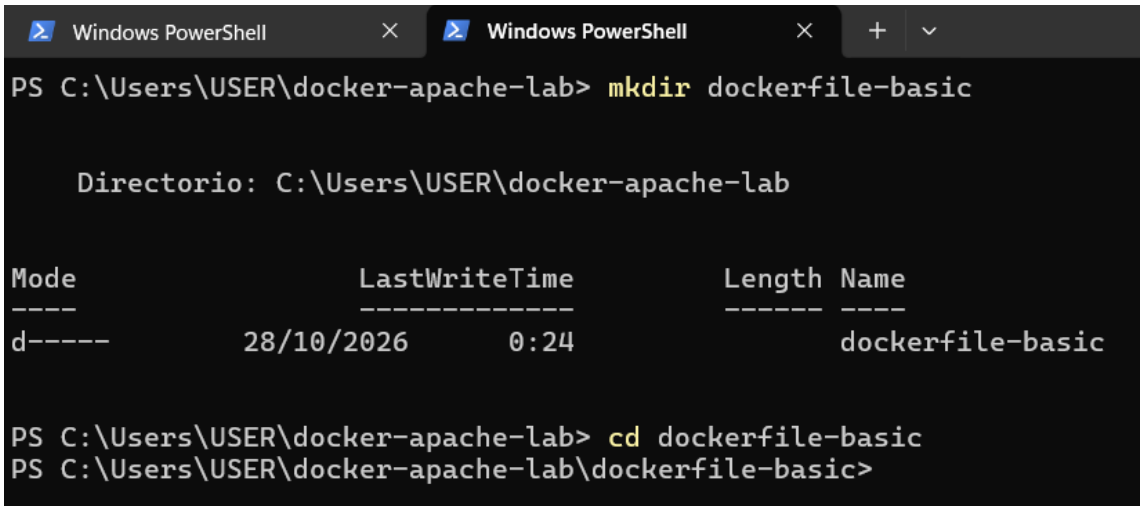
La pàgina Sobre mi es carrega correctament sense necessitat d'haver reiniciat el servei.



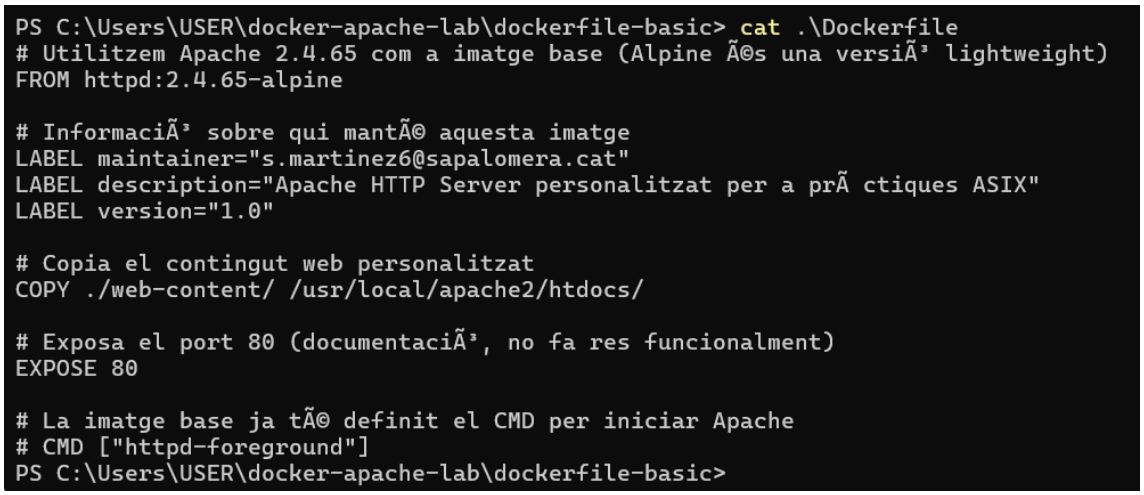
3. Part 2: Dockerfile – Creant les nostres pròpies imatges

El nostre primer Dockerfile

Creo el directori dockerfile-basic dins d'apache-lab.



Es mostra el fitxer de Dockerfile que defineix la imatge oficial (FROM...), també els LABEL que mostra el correu, la descripció i la versió. Amb COPY copia el contingut de la web personalitzada. Exposa el port 80.



Faig un ls per mostrar el contingut del directori (carpeta web-content creada).

cat .\web-content\index.html mostra la pàgina de docker personalitzada.

```
Windows PowerShell
PS C:\Users\USER\docker-apache-lab\dockerfile-basic> ls


    Directorio: C:\Users\USER\docker-apache-lab\dockerfile-basic


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        28/10/2026     15:19                web-content
-a----        28/10/2026     15:17            554 Dockerfile


PS C:\Users\USER\docker-apache-lab\dockerfile-basic> cat .\web-content\index.html
<!DOCTYPE html>
<html lang="ca">
<head>
    <meta charset="UTF-8">
    <title>Imatge Docker Personalitzada</title>
    <style>
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background: #2c3e50;
            color: #ecf0f1;
            padding: 50px;
            text-align: center;
        }
        .badge {
            background: #e74c3c;
            padding: 10px 20px;
            border-radius: 20px;
            display: inline-block;
            margin: 20px 0;
        }
    </style>
</head>
<body>
    <h1>ðŸš€ Imatge Docker Personalitzada</h1>
    <div class="badge">ConstruÃ¯da amb Dockerfile</div>
    <p>Aquesta pÃ gina s'ha copiat automÃ ticament durant la construcciÃ³ de la imatge!</p>
</body>
</html>
PS C:\Users\USER\docker-apache-lab\dockerfile-basic>
```

Construir la imatge

Utilitzo docker build per carregar el Dockerfile, mostro la nova creada. Executo el contenidor.

```
Windows PowerShell
PS C:\Users\USER\docker-apache-lab\dockerfile-basic> docker build -t apache-custom:v1.0 .
[+] Building 0.8s (7/7) FINISHED                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.0s
 => => transferring dockerfile: 593B                                                                                0.0s
 => [internal] load metadata for docker.io/library/httpd:2.4.65-alpine                                              0.5s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 2B                                                                                     0.0s
 => [internal] load build context                                                                                   0.0s
 => => transferring context: 75B                                                                                    0.0s
 => [1/2] FROM docker.io/library/httpd:2.4.65-alpine@sha256:07b2fabb7029a0b8aeb2e0fd02651c28fe22c21c5b5a59d6ff5b0   0.0s
 => => resolve docker.io/library/httpd:2.4.65-alpine@sha256:07b2fabb7029a0b8aeb2e0fd02651c28fe22c21c5b5a59d6ff5b0   0.0s
 => CACHED [2/2] COPY ./web-content/ /usr/local/apache2/htdocs/                                                     0.0s
 => exporting to image                                                                                              0.1s
 => => exporting layers                                                                                             0.0s
 => => exporting manifest sha256:e556e80fb8ba27995b06c641b55ba6f259491d324087f2b3382e2193e11e8871                   0.0s
 => => exporting config sha256:42db0d8e724f3a4da206a544d00935f5dcf3323bbcf3235dcf239a07db8143ac                     0.0s
 => => exporting attestation manifest sha256:aae80c6ac698586c55594610f94ed908b2e7ae409724c49e124db03a4b583116       0.0s
 => => exporting manifest list sha256:97ab052bd9351419c2f5a765f31b3ac00d2200bd9bee69351f4af18835568018              0.0s
 => => naming to docker.io/library/apache-custom:v1.0                                                               0.0s
 => => unpacking to docker.io/library/apache-custom:v1.0                                                            0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/f2ecnjmihwsgjvsy796ek0v9r
PS C:\Users\USER\docker-apache-lab\dockerfile-basic> docker images
                                                                                          i Info →  U In Use
IMAGE                  ID             DISK USAGE   CONTENT SIZE   EXTRA
apache-custom:v1.0     97ab052bd935       93.2MB         20.2MB
httpd:2.4.65           f9b88f3f093d        177MB         47.5MB    U
ubuntu/apache2:latest  bd68b3b35b01        228MB         52.2MB
PS C:\Users\USER\docker-apache-lab\dockerfile-basic> docker run -d -p 8081:80 --name apache-dockerfile apache-custom:v1.0
eed30171c94d8cd0de004aeb692364816fb7997bf01be9876490f837a4af9151
PS C:\Users\USER\docker-apache-lab\dockerfile-basic>
```

Resultat de la pàgina localhost:8081.



Multi-stage builds (construccions en múltiples etapes)

Creació del directori dockerfile-multistage.

```
PS C:\Users\USER\docker-apache-lab> mkdir dockerfile-multistage


    Directorio: C:\Users\USER\docker-apache-lab


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         28/10/2026     21:35                dockerfile-multistage


PS C:\Users\USER\docker-apache-lab> cd dockerfile-multistage
PS C:\Users\USER\docker-apache-lab\dockerfile-multistage>
```

Fitxer Dockerfile:

```
PS C:\Users\USER\docker-apache-lab> cat .\Dockerfile
# ETAPA 1: ConstrucciÃ³ del contingut
FROM node:18-alpine AS builder

# InstalÂ·la dependÃ¨ncies per "construir" el lloc web
WORKDIR /build

# En una aplicaciÃ³ real, aquÃ- compilaries TypeScript, minificaries CSS, etc.
# Per simplicitat, nomÃ©s crearem fitxers HTML
RUN mkdir -p dist && \
    echo '<!DOCTYPE html>' > dist/index.html && \
    echo '<html><head><title>Multi-stage Build</title></head>' >> dist/index.html && \
    echo '<body style="font-family: Arial; background: #344495e; color: white; padding: 50px; text-align: center;">' >> dist/index.html && \
    echo '<h1> Multi-Stage Build</h1>' >> dist/index.html && \
    echo '<p>Aquesta imatge s'\''ha creat utilitzant construcciÃ³ en mÃºltiples etapes</p>' >> dist/index.html && \
    echo '<p style="background: #e67e22; padding: 10px; border-radius: 5px;">Imatge final: nomÃ©s Apache + fitxers estÃ tics (sense Node.js)</p>' >> dist/in
dex.html && \
    echo '</body></html>' >> dist/index.html

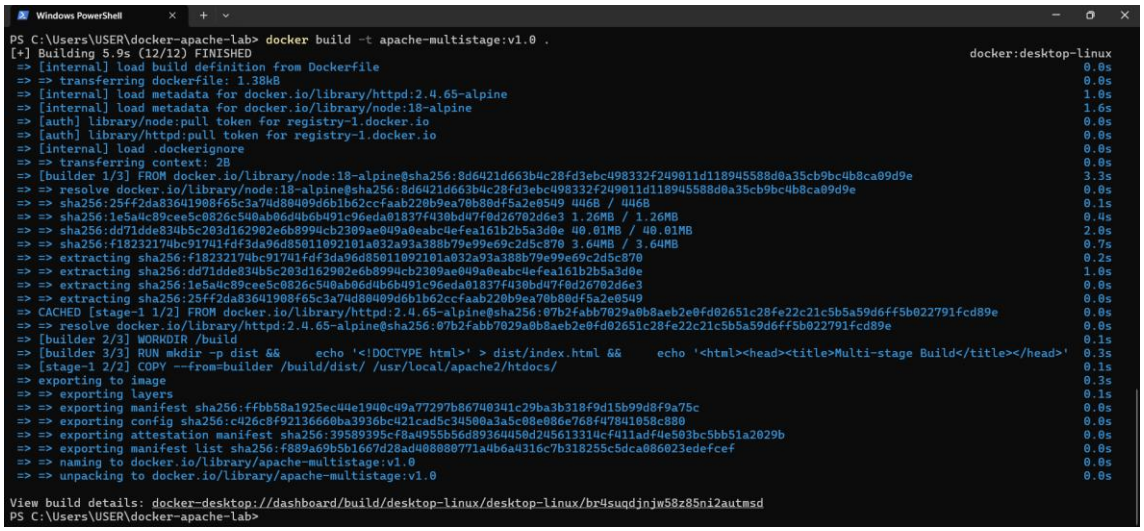
# ETAPA 2: Imatge final (nomÃ©s runtime)
FROM httpd:2.4.65-alpine

# Metadades
LABEL maintainer="s.martinez6@sapalomera.cat"
LABEL description="Apache amb multi-stage build"

# Copia NOMÃ©S els fitxers construÃ¯ts de l'etapa anterior
# No inclou Node.js ni les eines de construcciÃ³
COPY --from=builder /build/dist/ /usr/local/apache2/htdocs/

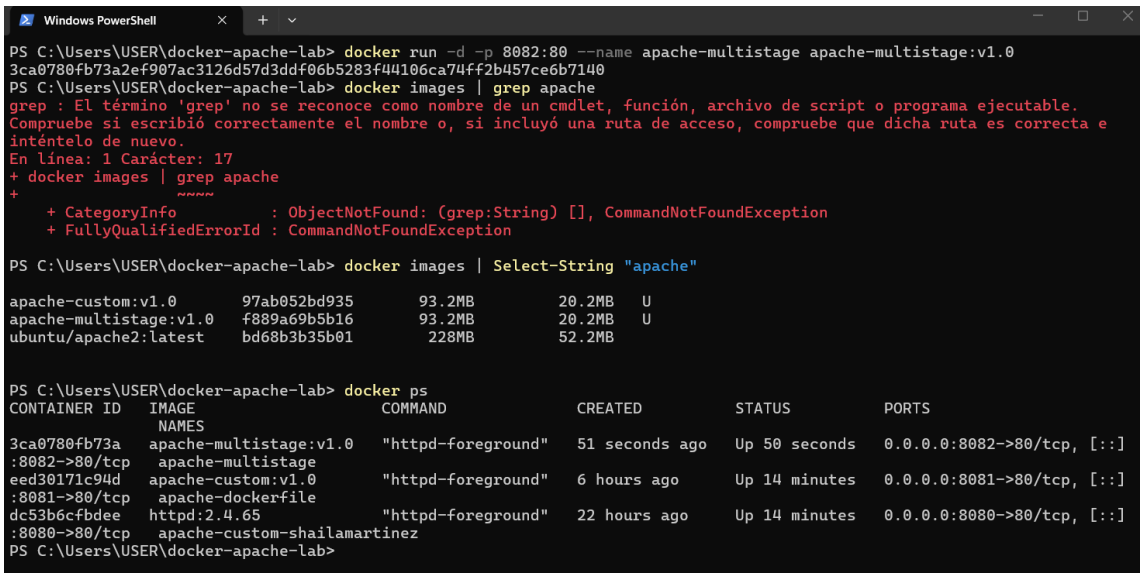
EXPOSE 80
PS C:\Users\USER\docker-apache-lab> docker build -t apache-multistage:v1.0 .
```

Build del apache-multistage acabat:

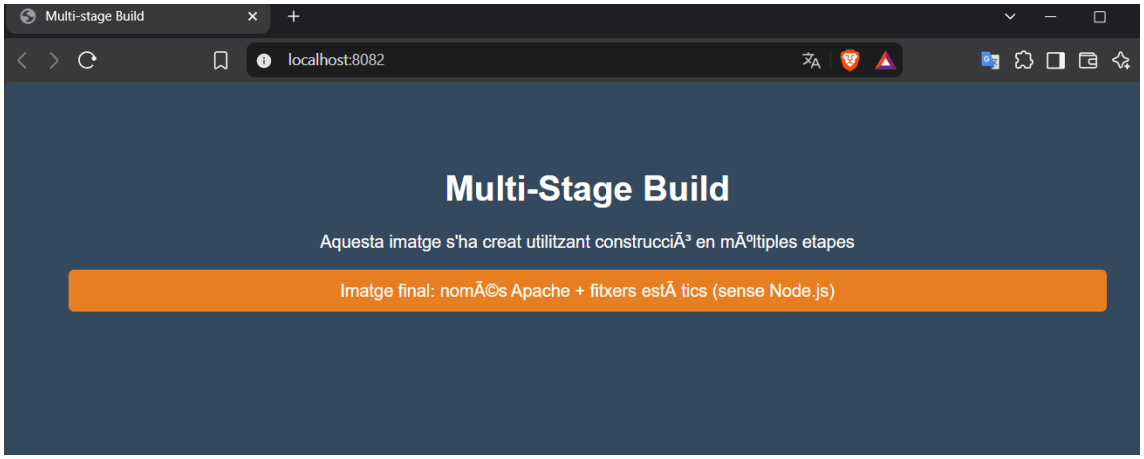


Arrenco el contenidor, mostra les versions del Apache custom i multistage. Docker ps surten els tres contenidors. Les imatges apache-custom i apache-multistage ocupen el mateix perquè a la imatge multi-stage només es guarda Apache amb Alpine i els fitxers HTML finals.

Node.js només s'utilitza durant la creació de la pàgina i no s'inclou a la imatge final, així queda una imatge més neta.



Resultat a http://localhost:8082



Best practices per a Dockerfile

Utilitzo una imatge ficial i lleugera amb versió concreta, exclusió de fitxers innecessaris amb .dockerignore, afegeio metadades amb Label i redueixo les capes. També deixo el Copy del web al final per aprofitar el cache. No he afegit User perquè Apache ja executa amb l'usuari daemon i no com root.

Repte 2

Estructura de carpetes i les 3 pàgines indicades dins de static-site.

```
PS C:\Users\USER\docker-apache-lab\dockerfile-repte2> ls


    Directorio: C:\Users\USER\docker-apache-lab\dockerfile-repte2


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        28/10/2026     21:56                static-site
-a----        28/10/2026     22:05             65 .dockerignore
-a----        28/10/2026     22:03            233 Dockerfile


PS C:\Users\USER\docker-apache-lab\dockerfile-repte2> ls .\static-site\


    Directorio: C:\Users\USER\docker-apache-lab\dockerfile-repte2\static-site


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        28/10/2026     21:57            314 contact.html
-a----        28/10/2026     21:54            266 index.html
-a----        28/10/2026     21:56            320 services.html


PS C:\Users\USER\docker-apache-lab\dockerfile-repte2>
```

Fitxer Dockerfile amb metadades a LABEL, contingut del dockerignore i index.html.

```
Windows PowerShell
PS C:\Users\USER\docker-apache-lab\dockerfile-repte2> cat .\Dockerfile
FROM httpd:2.4.65-alpine

LABEL maintainer="s.martinez6@sapalomera.cat"
LABEL description="Practiques Docker"
LABEL version="1.0"

WORKDIR /usr/local/apache2/htdocs

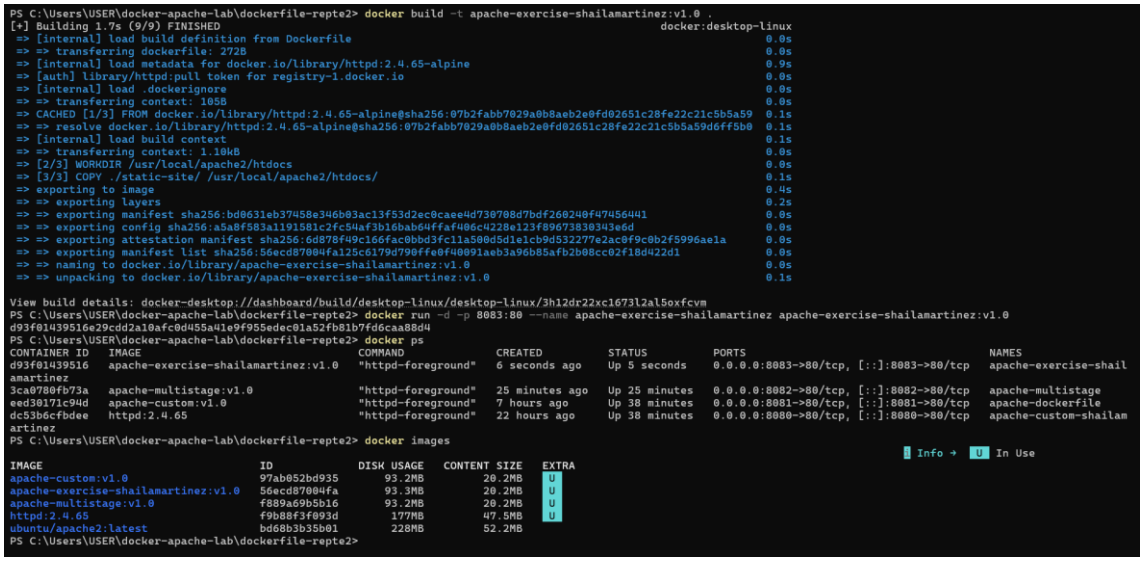
COPY ./static-site/ /usr/local/apache2/htdocs/

EXPOSE 80
PS C:\Users\USER\docker-apache-lab\dockerfile-repte2> cat .\.dockerignore
.git
.gitignore
README.md
node_modules
*.log
.env
.DS_Store
PS C:\Users\USER\docker-apache-lab\dockerfile-repte2> cat .\static-site\index.html
<!DOCTYPE html>
<html lang="ca">
<head>
    <meta charset="UTF-8">
    <title>index</title>
</head>
<body>
    <h1>Inici web</h1>
    <p>2n ASIX Pr.Docker</p>
    <a href="services.html">Serveis</a>
    <a href="contact.html">Contacte</a>
</body>
</html>
PS C:\Users\USER\docker-apache-lab\dockerfile-repte2> cat .\static-site\services.html
```

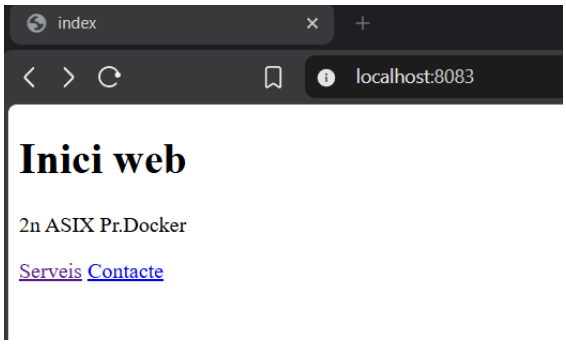
Codi de la pàgina services.html, contact.html

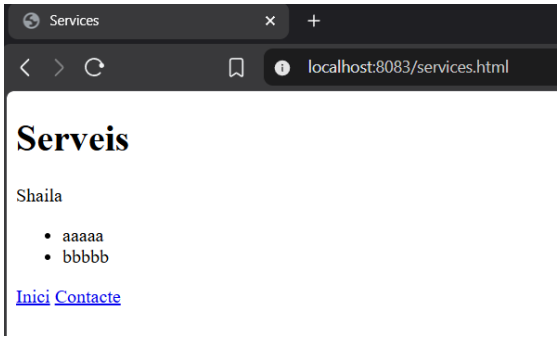
```
PS C:\Users\USER\docker-apache-lab\dockerfile-repte2> cat .\static-site\services.html
<!DOCTYPE html>
<html lang="ca">
<head>
    <meta charset="UTF-8">
    <title>Services</title>
</head>
<body>
    <h1>Serveis</h1>
    <p>Shaila</p>
    <ul>
        <li>aaaaa</li>
        <li>bbbbb</li>
    </ul>
    <a href="index.html">Inici</a>
    <a href="contact.html">Contacte</a>
</body>
</html>
PS C:\Users\USER\docker-apache-lab\dockerfile-repte2> cat .\static-site\contact.html
<!DOCTYPE html>
<html lang="ca">
<head>
    <meta charset="UTF-8">
    <title>Contacte</title>
</head>
<body>
    <h1>Contacte</h1>
    <p>Nom: Shaila MartÃ-nez</p>
    <p>Correu: s.martinez6@sapalomera.cat</p>
    <a href="index.html">Inici</a>
    <a href="services.html">Serveis</a>
</body>
</html>
PS C:\Users\USER\docker-apache-lab\dockerfile-repte2>
```

Construcció i execució amb docker buid i run. Es veuen els 4 contenidors actius i la nova imatge.

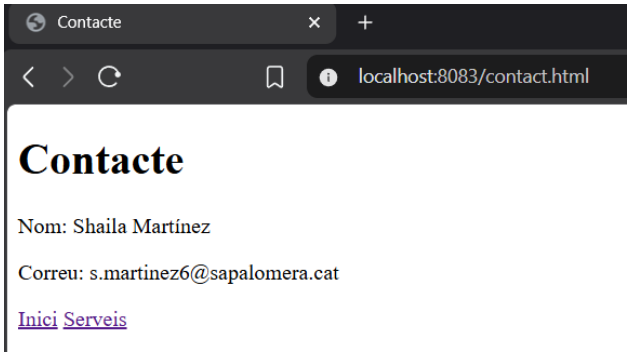


Localhost:8083 mostra l'inici web, services.html serveis.





Pàgina de contacte:



4. Part 3: Apache HTTP Server – Configuració bàsica

Mòduls d'Apache

Verificació dels mòduls carregats i contenidors en marxa. A la llista no hi apareixen rewrite_module ni ssl_module perquè la imatge oficial no les carrega per defecte. Quan es necessiten es carreguen amb LoadModule (a la configuració pròpia).

```
PS C:\Users\USER\docker-apache-lab\dockerfile-repte2> docker exec apache-custom-shailamartinez httpd -M
AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using 172.17.0.3. Set the 'ServerName' directive globally to suppress
 this message
Loaded Modules:
 core_module (static)
 so_module (static)
 http_module (static)
 mpm_event_module (shared)
 authn_file_module (shared)
 authn_core_module (shared)
 authz_host_module (shared)
 authz_groupfile_module (shared)
 authz_user_module (shared)
 authz_core_module (shared)
 access_compat_module (shared)
 auth_basic_module (shared)
 reqtimeout_module (shared)
 filter_module (shared)
 mime_module (shared)
 log_config_module (shared)
 env_module (shared)
 headers_module (shared)
 setenvif_module (shared)
 version_module (shared)
 unixd_module (shared)
 status_module (shared)
 autoindex_module (shared)
 dir_module (shared)
 alias_module (shared)
PS C:\Users\USER\docker-apache-lab\dockerfile-repte2> docker ps -a
CONTAINER ID   IMAGE                              COMMAND              CREATED          STATUS          PORTS                                       NAMES
d93f01439516   apache-exercise-shailamartinez:v1.0   "httpd-foreground"   6 minutes ago    Up 6 minutes    0.0.0.0:8083->80/tcp, [::]:8083->80/tcp   apache-
exercise-shailamartinez
3ca0780fb73a   apache-multistage:v1.0             "httpd-foreground"   31 minutes ago   Up 31 minutes   0.0.0.0:8082->80/tcp, [::]:8082->80/tcp   apache-
multistage
eed30171c94d   apache-custom:v1.0                 "httpd-foreground"   7 hours ago      Up 45 minutes   0.0.0.0:8081->80/tcp, [::]:8081->80/tcp   apache-
dockerfile
dc53b6cfbdee   httpd:2.4.65                       "httpd-foreground"   22 hours ago     Up 45 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   apache-
custom-shailamartinez
PS C:\Users\USER\docker-apache-lab\dockerfile-repte2>
```

Configuració personalitzada d'Apache

Estructura de apache-config i contingut de Dockerfile.

```
Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        28/10/2026     22:20                conf
d-----        28/10/2026     22:22                html
d-----        28/10/2026     22:20                logs
-a----        28/10/2026     22:22            349 Dockerfile


PS C:\Users\USER\docker-apache-lab\apache-config> tree /F
Listado de rutas de carpetas
El número de serie del volumen es 38C8-F55B
C:.
│   Dockerfile
│
├───conf
│       httpd-custom.conf
│
├───html
│       index.html
│
└───logs
PS C:\Users\USER\docker-apache-lab\apache-config> cat .\Dockerfile
FROM httpd:2.4.65-alpine

LABEL maintainer="s.martinez6@sapalomera.cat"
LABEL description="Apache amb configuraciÃ³ personalitzada"

# Copia la configuraciÃ³ personalitzada
COPY conf/httpd-custom.conf /usr/local/apache2/conf/httpd.conf

# Copia el contingut web
COPY html/ /usr/local/apache2/htdocs/

EXPOSE 80

CMD ["httpd-foreground"]
PS C:\Users\USER\docker-apache-lab\apache-config>
```

La configuració (httpd-custom.conf) carrega els mòduls, mpm és obligatori perquè sense ell Apache no arrenca amb l'error No MPM loaded. Permet mostrar els

fitxers del directori, seguir enllaços i utilitzar .htaccess. També configura els logs per guardar les peticions i el temps de resposta, capçaleres de seguretat...

```
PS C:\Users\USER\docker-apache-lab\apache-config> cat .\conf\httpd-custom.conf
# Carrega els mÃ²duls necessaris
# L'MPM Ã©s obligatori: sense ell Apache no arrenca ("No MPM loaded")
LoadModule mpm_event_module modules/mod_mpm_event.so
LoadModule authz_core_module modules/mod_authz_core.so
LoadModule dir_module modules/mod_dir.so
LoadModule mime_module modules/mod_mime.so
LoadModule log_config_module modules/mod_log_config.so
LoadModule unixd_module modules/mod_unixd.so
LoadModule rewrite_module modules/mod_rewrite.so
LoadModule headers_module modules/mod_headers.so

# ConfiguraciÃ³ del servidor
ServerName localhost
Listen 80

# Usuari i grup (per defecte en la imatge oficial)
User daemon
Group daemon

# Document Root
DocumentRoot "/usr/local/apache2/htdocs"

<Directory "/usr/local/apache2/htdocs">
    # Permet seguir enllaÃ§os simbÃ²lics
    Options Indexes FollowSymLinks

    # Permet l'Ãºs de .htaccess
    AllowOverride All

    # Permet accÃ©s a tothom
    Require all granted

    # Activa el mod_rewrite per aquest directori
    RewriteEngine On
</Directory>

# Index per defecte
DirectoryIndex index.html index.htm

# Tipus MIME
TypesConfig conf/mime.types

# ConfiguraciÃ³ de logging
ErrorLog "logs/error.log"
LogLevel warn

# Format de log personalitzat (Combined Log Format + temps de resposta)
LogFormat "%h %l %u %t \"%r\" %>s %b \"%{Referer}i\" \"%{User-Agent}i\" %D" combined_with_time
CustomLog "logs/access.log" combined_with_time

# CapÃ§alera de seguretat (exemple)
Header always set X-Content-Type-Options "nosniff"
Header always set X-Frame-Options "SAMEORIGIN"
PS C:\Users\USER\docker-apache-lab\apache-config>
```

Index.html amb l'estil:

```
PS C:\Users\USER\docker-apache-lab\apache-config> cat .\html\index.html
<!DOCTYPE html>
<html lang="ca">
<head>
    <meta charset="UTF-8">
    <title>Apache Configurat</title>
    <style>
        body {
            font-family: 'Courier New', monospace;
            background: #1a1a2e;
            color: #eee;
            padding: 40px;
        }
        .config-box {
            background: #16213e;
            border-left: 5px solid #0f3460;
            padding: 20px;
            margin: 20px 0;
            border-radius: 5px;
        }
        code {
            background: #0f3460;
            padding: 2px 8px;
            border-radius: 3px;
            color: #feca57;
        }
    </style>
</head>
<body>
    <h1>Apache amb ConfiguraciÃ³ Personalitzada</h1>

    <div class="config-box">
        <h3>Funcionalitats actives:</h3>
        <ul>
            <li>MÃ²dul Rewrite activat</li>
            <li>CapÃ§aleres de seguretat configurades</li>
            <li>Logging personalitzat amb temps de resposta</li>
            <li>.htaccess permÃ¨s (AllowOverride All)</li>
        </ul>
    </div>

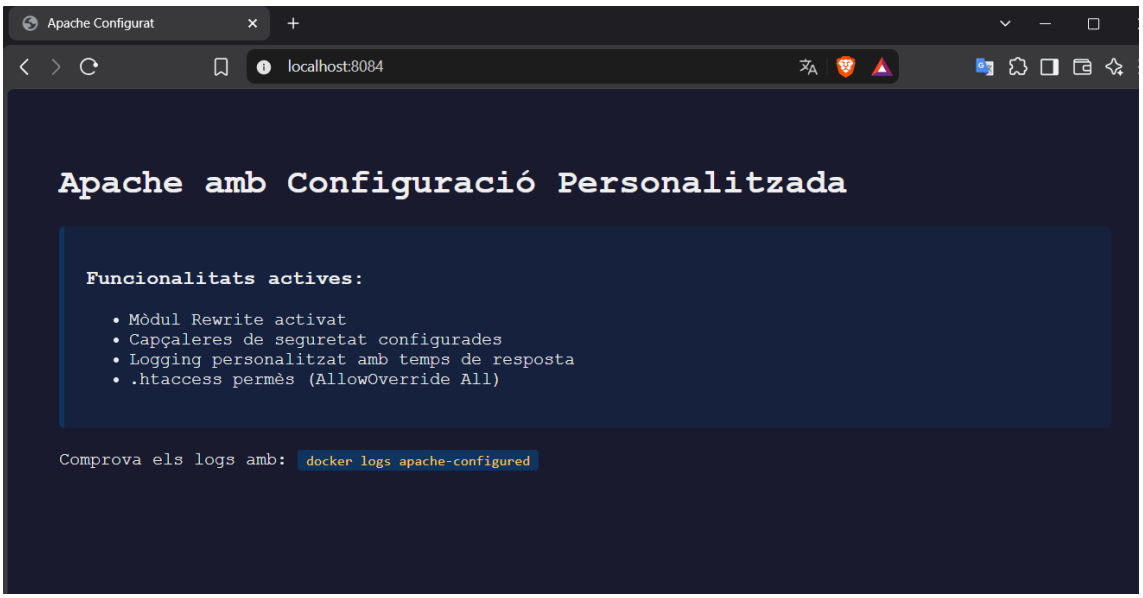
    <p>Comprova els logs amb: <code>docker logs apache-configured</code></p>
</body>
</html>
PS C:\Users\USER\docker-apache-lab\apache-config>
```

Construcció i execució:

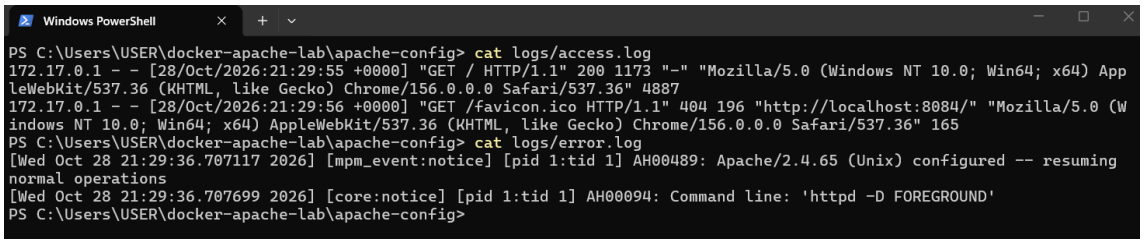
```
PS C:\Users\USER\docker-apache-lab\apache-config> docker build -t apache-configured:v1.0 .
[+] Building 1.7s (9/9) FINISHED                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                0.0s
 => => transferring dockerfile: 388B                                                                0.0s
 => [internal] load metadata for docker.io/library/httpd:2.4.65-alpine                              0.9s
 => [auth] library/httpd:pull token for registry-1.docker.io                                        0.0s
 => [internal] load .dockerignore                                                                   0.0s
 => => transferring context: 2B                                                                     0.0s
 => CACHED [1/3] FROM docker.io/library/httpd:2.4.65-alpine@sha256:07b2fabb7029a0b8aeb2e0fd02651c28fe22c21c5b5a59  0.0s
 => => resolve docker.io/library/httpd:2.4.65-alpine@sha256:07b2fabb7029a0b8aeb2e0fd02651c28fe22c21c5b5a59d6ff5b0  0.0s
 => [internal] load build context                                                                   0.1s
 => => transferring context: 2.89kB                                                                 0.0s
 => [2/3] COPY conf/httpd-custom.conf /usr/local/apache2/conf/httpd.conf                            0.0s
 => [3/3] COPY html/ /usr/local/apache2/htdocs/                                                     0.1s
 => exporting to image                                                                              0.4s
 => => exporting layers                                                                             0.2s
 => => exporting manifest sha256:a38b277f3e21e7884f92cba5d569fc64626deb66eb421c3b31d8e333f6584cce   0.0s
 => => exporting config sha256:503f1d4205bf172f4166ab887b01de00bcf6cb895b14cd3b63a53fc7db489044     0.0s
 => => exporting attestation manifest sha256:1c14d45dd0cc98a5ea2bbcb43cbb14bbb5f23d9937221e7700c2912a348e917c  0.0s
 => => exporting manifest list sha256:1996d93416acabe08596a4e084cb28bc7edb5d8ee09c7029084b2045bcd81584  0.0s
 => => naming to docker.io/library/apache-configured:v1.0                                           0.0s
 => => unpacking to docker.io/library/apache-configured:v1.0                                        0.1s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/fw1z81sbcqp0bzxxm1nwalgm4
PS C:\Users\USER\docker-apache-lab\apache-config> docker run -d -p 8084:80 --name apache-configured -v "${PWD}/logs:/usr
/local/apache2/logs" apache-configured:v1.0
8b138aae17df09d9e38e4736489330afad9d261acf4bfaa01e38edfe79da944a
PS C:\Users\USER\docker-apache-lab\apache-config>
```

Resultat http://localhost:8084. Mostra la pàgina Apache amb Configuració Personalitzada amb la llista de funcionalitats.



Logs d'accés i d'error, mostra les peticions amb el format i el temps de resposta.



Fitxers .htaccess

Creació del fitxer .htaccess a html\ : URL sense extensió, protegir fitxers de configuració, capçalera personalitzada per utilitzar .htaccess i Options -Indexes per desactivar el llistat de directoris. Després s'atura i s'elimina el contenidor.

```
Windows PowerShell
PS C:\Users\USER\docker-apache-lab\apache-config> ls .\html\


    Directorio: C:\Users\USER\docker-apache-lab\apache-config\html


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        29/10/2026     12:25            497 .htaccess
-a----        28/10/2026     22:22           1173 index.html


PS C:\Users\USER\docker-apache-lab\apache-config> cat .\html\.htaccess
# Activar el motor de rewrite
RewriteEngine On

# Exemple: URLs sense extensiÃ³ (/pagina serveix /pagina.html si existeix)
RewriteCond %{REQUEST_FILENAME}.html -f
RewriteRule ^(.*)$ $1.html [L]

# Protegir fitxers de configuraciÃ³
<FilesMatch "^\.">
    Require all denied
</FilesMatch>

# CapÃ§alera personalitzada per indicar que s'estÃ  utilitzant .htaccess
Header set X-Powered-By "Apache-ASIX-Lab"

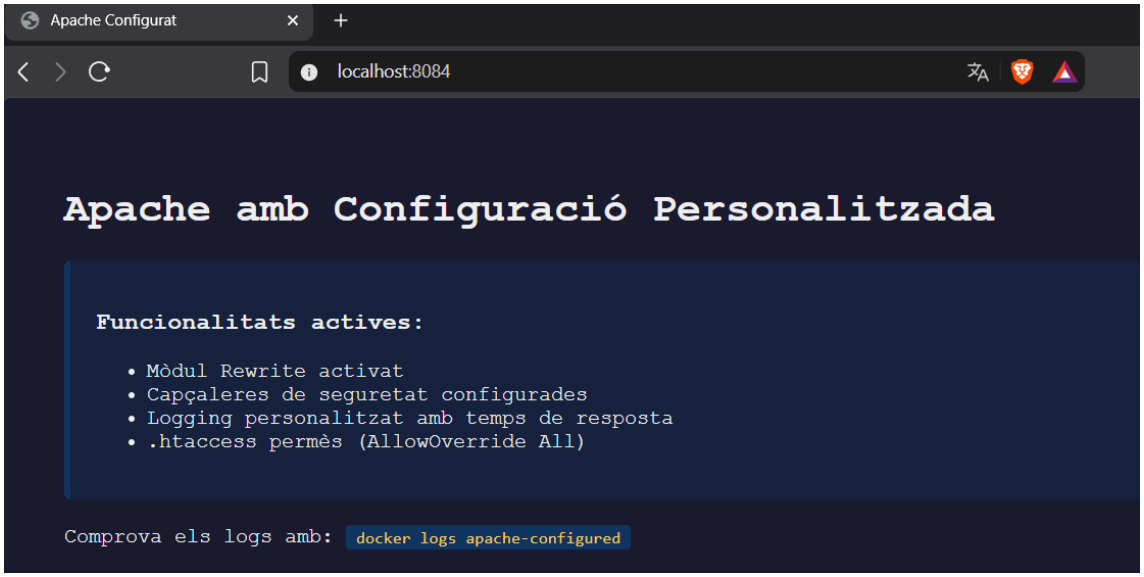
# Desactivar listatge de directoris (si no hi ha index.html)
Options -Indexes
PS C:\Users\USER\docker-apache-lab\apache-config> docker stop apache-configured
apache-configured
PS C:\Users\USER\docker-apache-lab\apache-config> docker rm apache-configured
apache-configured
PS C:\Users\USER\docker-apache-lab\apache-config>
```

Reconstrueixo la imatge i torno a executar el contenidor al port 8084 amb el volum de logs.

```
Windows PowerShell
PS C:\Users\USER\docker-apache-lab\apache-config> docker build -t apache-configured:v1.1 .
[+] Building 3.0s (9/9) FINISHED                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.1s
 => => transferring dockerfile: 388B                                                                                0.0s
 => [internal] load metadata for docker.io/library/httpd:2.4.65-alpine                                              1.4s
 => [auth] library/httpd:pull token for registry-1.docker.io                                                        0.0s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 2B                                                                                     0.0s
 => [1/3] FROM docker.io/library/httpd:2.4.65-alpine@sha256:07b2fabb7029a0b8aeb2e0fd02651c28fe22c21c5b5a59d6ff5b0  0.1s
 => => resolve docker.io/library/httpd:2.4.65-alpine@sha256:07b2fabb7029a0b8aeb2e0fd02651c28fe22c21c5b5a59d6ff5b0  0.1s
 => [internal] load build context                                                                                   0.1s
 => => transferring context: 669B                                                                                   0.0s
 => CACHED [2/3] COPY conf/httpd-custom.conf /usr/local/apache2/conf/httpd.conf                                     0.0s
 => [3/3] COPY html/ /usr/local/apache2/htdocs/                                                                     0.2s
 => exporting to image                                                                                              0.9s
 => => exporting layers                                                                                             0.3s
 => => exporting manifest sha256:5af50d4274841bae3bcb4ba58b9a02847fa3aa47427b246d7a1b82b17ad5c5b0                    0.1s
 => => exporting config sha256:d80b7ae90364f531439539c49c5ca04273d4e16f41a28d91238a7b2317b7bf5c                      0.1s
 => => exporting attestation manifest sha256:c145da076577e74df37487a5c24f4703ab168033e977f2f770d135d5b8de6af2       0.1s
 => => exporting manifest list sha256:5e08868561b09a40ef4a5a9ae7fe25e2d9d64167f4d55256dc3688244436c693              0.1s
 => => naming to docker.io/library/apache-configured:v1.1                                                           0.0s
 => => unpacking to docker.io/library/apache-configured:v1.1                                                        0.1s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/uz6rhlx2l7v4eb6cbg9cjrju9
PS C:\Users\USER\docker-apache-lab\apache-config> docker run -d -p 8084:80 --name apache-configured -v "$(pwd)/logs:/usr
/local/apache2/logs" apache-configured:v1.1
ad2586b12b372cd1a096b7910707e8ec779753697fb24723258f5b9c83909ad9
PS C:\Users\USER\docker-apache-lab\apache-config>
```

Verificació a localhost:8084 la pàgina continua carregant-se correctament amb la nova imatge.



Repte 3

Afegeixo una regla perquè quan algú entri a /old-page es mostri el contingut de /new-page.html. També afegeixo una capçalera que indica la versió del servidor a totes les respostes.

```
PS C:\Users\USER\docker-apache-lab\apache-config> cat .\html\.htaccess
RewriteEngine On

RewriteCond %{REQUEST_FILENAME}.html -f
RewriteRule ^(.*)$ $1.html [L]

RewriteRule ^old-page$ /new-page.html [L]

<FilesMatch "^\.">
    Require all denied
</FilesMatch>

Header set X-Powered-By "Apache-ASIX-Lab"
Header always set X-Lab-Version "1.0"

Options -Indexes
PS C:\Users\USER\docker-apache-lab\apache-config>
```

Creo new-page.html i faig ls per veure els 3 fitxers dins de /html/.

```
PS C:\Users\USER\docker-apache-lab\apache-config> cat .\html\new-page.html
<!DOCTYPE html>
<html lang="ca">
<head>
    <meta charset="UTF-8">
    <title>Nova pag</title>
</head>
<body>
    <h1>Nova pÃ gina</h1>
    <p>.htaccess</p>
</body>
</html>
PS C:\Users\USER\docker-apache-lab\apache-config> ls .\html\


    Directorio: C:\Users\USER\docker-apache-lab\apache-config\html


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         29/10/2026     15:26            301 .htaccess
-a----         28/10/2026     22:22           1173 index.html
-a----         29/10/2026     15:29            182 new-page.html

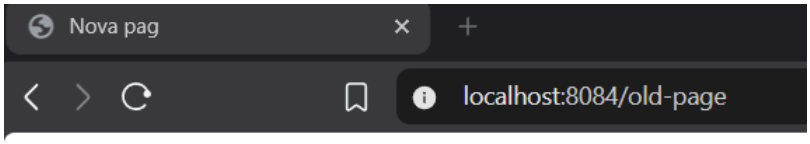

PS C:\Users\USER\docker-apache-lab\apache-config>
```

Repeteixo el procés per reconstruir la imatge al port 8084.

```
PS C:\Users\USER\docker-apache-lab\apache-config> docker stop apache-configured
apache-configured
PS C:\Users\USER\docker-apache-lab\apache-config> docker rm apache-configured
apache-configured
PS C:\Users\USER\docker-apache-lab\apache-config> docker build -t apache-configured:v1.2 .
[+] Building 1.8s (9/9) FINISHED                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                              0.0s
 => => transferring dockerfile: 388B                                                                                              0.0s
 => [internal] load metadata for docker.io/library/httpd:2.4.65-alpine                                                            1.1s
 => [auth] library/httpd:pull token for registry-1.docker.io                                                                      0.0s
 => [internal] load .dockerignore                                                                                                 0.0s
 => => transferring context: 2B                                                                                                   0.0s
 => [1/3] FROM docker.io/library/httpd:2.4.65-alpine@sha256:07b2fabb7029a0b8aeb2e0fd02651c28fe22c21c5b5a59d6ff5b0                 0.0s
 => => resolve docker.io/library/httpd:2.4.65-alpine@sha256:07b2fabb7029a0b8aeb2e0fd02651c28fe22c21c5b5a59d6ff5b0                 0.0s
 => [internal] load build context                                                                                                 0.0s
 => => transferring context: 704B                                                                                                 0.0s
 => CACHED [2/3] COPY conf/httpd-custom.conf /usr/local/apache2/conf/httpd.conf                                                   0.0s
 => [3/3] COPY html/ /usr/local/apache2/htdocs/                                                                                   0.1s
 => exporting to image                                                                                                            0.4s
 => => exporting layers                                                                                                           0.1s
 => => exporting manifest sha256:dfa1f88388295c37c5372e35d13e9a4cf31bf7155fc57eb9f1ce7e0b9314fd00                                 0.0s
 => => exporting config sha256:f7edc0086edf795e2055cb0edd907aa91d925e6bb69bda8f7e3fea30d12a5530                                   0.0s
 => => exporting attestation manifest sha256:7aa9a12994e809d0c777d2ce0442a04e398075f7c3b5bd9e825dc5295f48a40e                     0.0s
 => => exporting manifest list sha256:7757f543dcb60050bb7762e4def0f79c5d67caa5a95ec7fba8b5ae4d62378bdc                            0.0s
 => => naming to docker.io/library/apache-configured:v1.2                                                                         0.0s
 => => unpacking to docker.io/library/apache-configured:v1.2                                                                      0.1s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/2aswniyvhy8w02pgwua4wmhuv
PS C:\Users\USER\docker-apache-lab\apache-config> docker run -d -p 8084:80 --name apache-configured -v "${PWD}/logs:/usr/local/apache2/logs" apache-configur
ed:v1.2
8c002aa4369bd4a0004e581e9fd77bb518b43b053bd833a782d9e20ad10c1bb8
PS C:\Users\USER\docker-apache-lab\apache-config>
```

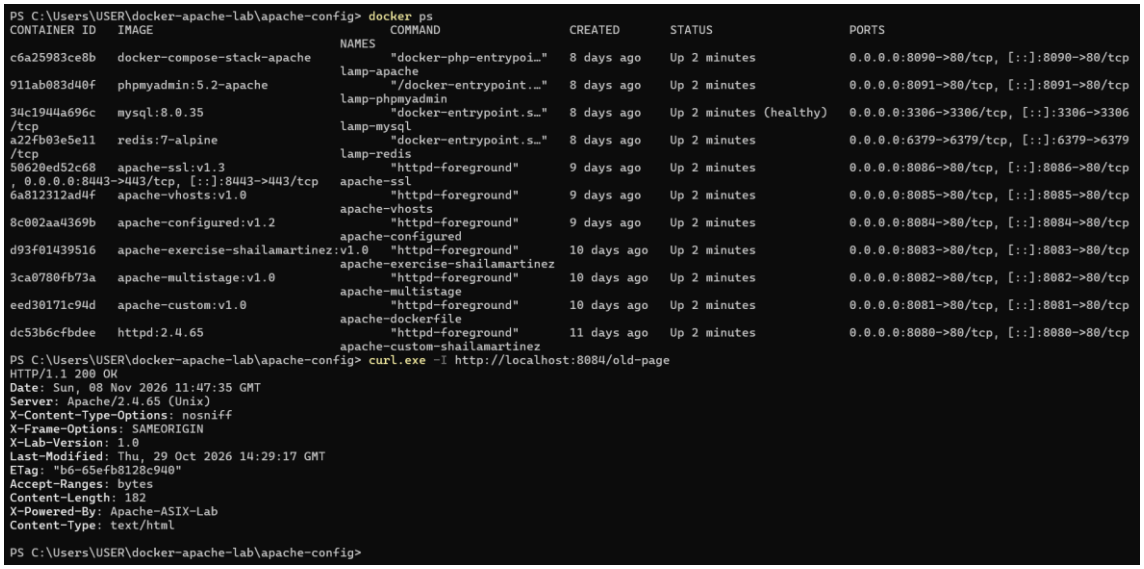
La URL queda a old-page però es mostra el contingut de new-page.html. Com que la regla usa [L] sense R, és una reescriptura interna i no una redirecció HTTP.



Nova pàgina

.htaccess

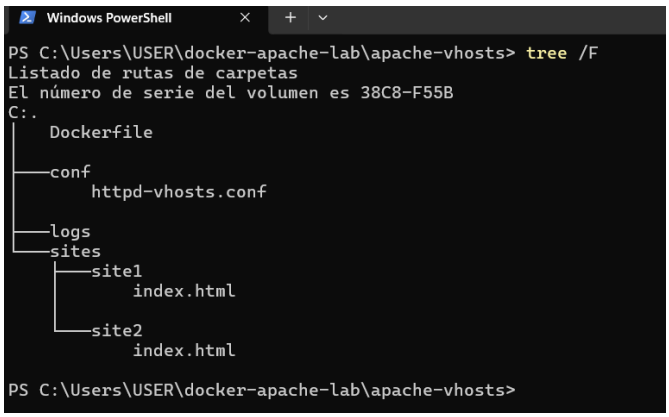
Comprovo que /old-page funciona correctament i que la resposta inclou la capçalera X-Lab-Version: 1.0



5. Part 4: Configuració avançada d'Apache

Virtual Hosts - Múltiples llocs web

Estructura de apache-vhosts.



httpd-vhosts.conf carrega els mòduls necessaris, defineix ServerName localhost, Listen 80 i un format de log vhost_combined. Té tres blocs VirtualHost *:80

- Un per defecte que atén les peticions que no coincideixen amb cap altre.
- site1.local (amb ServerAlias www.site1.local).
- site2.local amb la mateixa estructura



Contingut dels dos llocs:



Index.html del site 1 i 2:

```
PS C:\Users\USER\docker-apache-lab\apache-vhosts> cat .\sites\site1\index.html
<!DOCTYPE html>
<html lang="ca">
<head>
    <meta charset="UTF-8">
    <title>Lloc 1 - Empresa Tech</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            color: white;
            padding: 50px;
            text-align: center;
        }
        .logo { font-size: 4em; }
    </style>
</head>
<body>
    <div class="logo">ðŸš€</div>
    <h1>Empresa Tech Solutions</h1>
    <p>Virtual Host 1: site1.local</p>
    <p>Desenvolupament de software i consultoria IT</p>
</body>
</html>
PS C:\Users\USER\docker-apache-lab\apache-vhosts> cat .\sites\site2\index.html
<!DOCTYPE html>
<html lang="ca">
<head>
    <meta charset="UTF-8">
    <title>Lloc 2 - Disseny Web</title>
    <style>
        body {
            font-family: 'Georgia', serif;
            background: linear-gradient(135deg, #f093fb 0%, #f5576c 100%);
            color: white;
            padding: 50px;
            text-align: center;
        }
        .logo { font-size: 4em; }
    </style>
</head>
<body>
    <div class="logo">ðŸŽ¨</div>
    <h1>Creative Design Studio</h1>
    <p>Virtual Host 2: site2.local</p>
    <p>Disseny grÃ fic i experiÃ¨ncia d'usuari</p>
</body>
</html>
PS C:\Users\USER\docker-apache-lab\apache-vhosts>
```

Fitxer Dockerfile copia conf/httpd-vhosts.conf com a httpd.conf i sites/ a /usr/local/apache2/sites/

```
PS C:\Users\USER\docker-apache-lab\apache-vhosts> cat .\Dockerfile
FROM httpd:2.4.65-alpine

LABEL maintainer="s.martinez6@sapalomera.cat"
LABEL description="Apache amb Virtual Hosts"

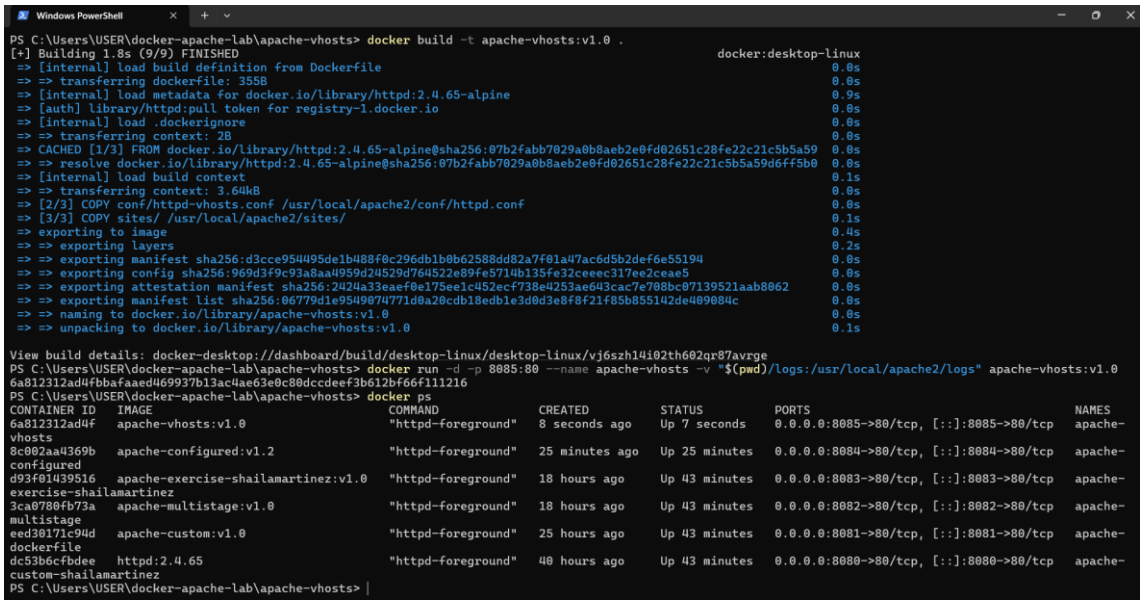
# Copia la configuraciÃ³
COPY conf/httpd-vhosts.conf /usr/local/apache2/conf/httpd.conf

# Copia els llocs web
COPY sites/ /usr/local/apache2/sites/

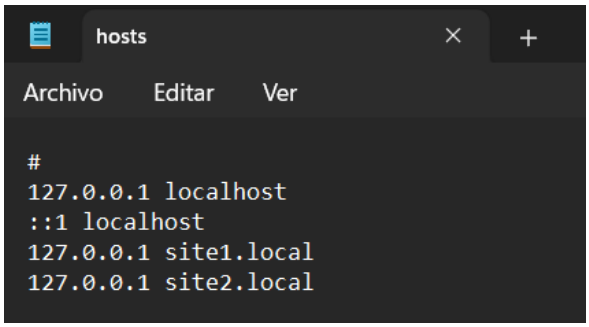
EXPOSE 80

CMD ["httpd-foreground"]
PS C:\Users\USER\docker-apache-lab\apache-vhosts>
```

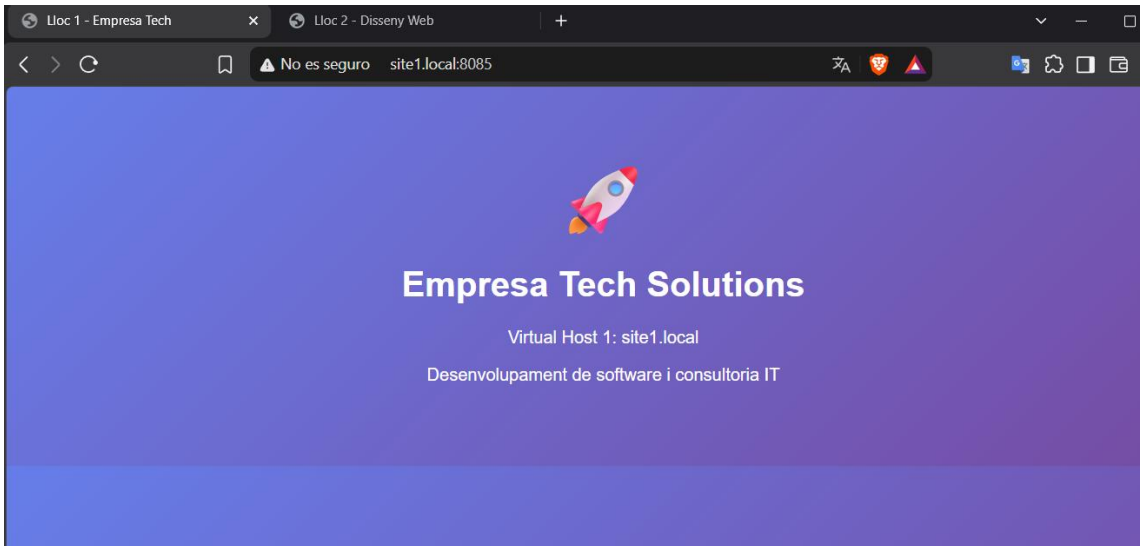
Execució i els contenidors (8080 - 8085):



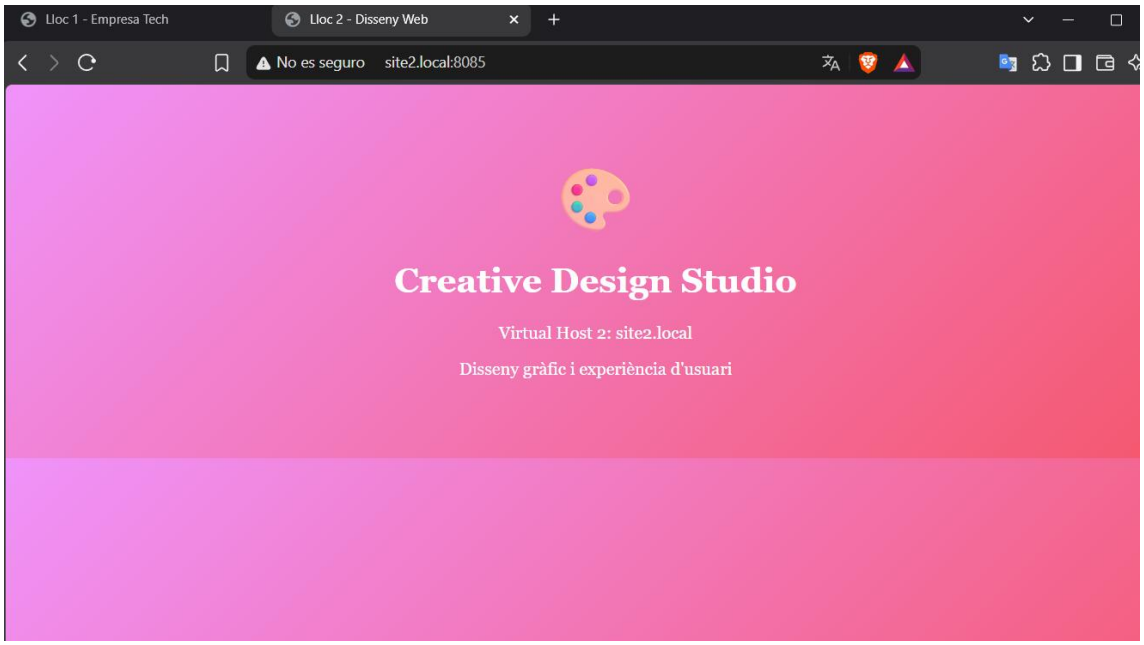
Fitxer hosts per resoldre els dominis locals, he afegit 127.0.0.1 site1.local i site2.local



Resultat al navegador :8085. Amb la mateixa IP i port es serveixen dos llocs diferents.



site2.local



Els logs mostren les peticions de cada lloc amb el nom de virtual host.

```
PS C:\Users\USER\docker-apache-lab\apache-vhosts> Get-Content .\logs\site1-access.log
site1.local 172.17.0.1 - - [29/Oct/2026:15:29:58 +0000] "GET / HTTP/1.1" 200 612
site1.local 172.17.0.1 - - [29/Oct/2026:15:29:58 +0000] "GET /favicon.ico HTTP/1.1" 404 196
PS C:\Users\USER\docker-apache-lab\apache-vhosts> Get-Content .\logs\site2-access.log
site2.local 172.17.0.1 - - [29/Oct/2026:15:30:10 +0000] "GET / HTTP/1.1" 200 605
site2.local 172.17.0.1 - - [29/Oct/2026:15:30:10 +0000] "GET /favicon.ico HTTP/1.1" 404 196
PS C:\Users\USER\docker-apache-lab\apache-vhosts>
```

HTTPS amb SSL/TLS

Estructura i Dockerfile de apache-ssl (generar certificats).

```
PS C:\Users\USER\docker-apache-lab\apache-ssl> tree /F
Listado de rutas de carpetas
El número de serie del volumen es 38C8-F55B
C:.
│   Dockerfile
│
├───certs
├───conf
│       httpd-ssl.conf
│
├───html
│       index.html
│
└───logs
PS C:\Users\USER\docker-apache-lab\apache-ssl> cat .\Dockerfile
FROM httpd:2.4.65-alpine

LABEL maintainer="s.martinez6@sapalomera.cat"
LABEL description="Apache amb HTTPS/SSL"

# InstalÂ·la OpenSSL (ja ve a Alpine, perÃ² assegurem dependÃ¨ncies)
RUN apk add --no-cache openssl

# Crea el directori per als certificats
RUN mkdir -p /usr/local/apache2/certs

# Genera un certificat auto-signat
# AtenciÃ³: AixÃ² Ã©s NOMÃ‰S per a desenvolupament! En producciÃ³ usa Let's Encrypt o un CA oficial
RUN openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
    -keyout /usr/local/apache2/certs/server.key \
    -out /usr/local/apache2/certs/server.crt \
    -subj "/C=ES/ST=Catalunya/L=Barcelona/O=ASIX Lab/OU=IT/CN=localhost"

# Ajusta permisos dels certificats
RUN chmod 600 /usr/local/apache2/certs/server.key && \
    chmod 644 /usr/local/apache2/certs/server.crt

# Copia la configuraciÃ³ d'Apache amb SSL
COPY conf/httpd-ssl.conf /usr/local/apache2/conf/httpd.conf

# Copia el contingut web
COPY html/ /usr/local/apache2/htdocs/

# Exposa els ports HTTP i HTTPS
EXPOSE 80 443

CMD ["httpd-foreground"]
PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

Configuració SSL, defineix el logging global, la cache de sessions SSL... Inclou VirtualHost 443 amb SSLEngine on, les rutes al certificat i la clau.

```
PS C:\Users\USER\docker-apache-lab\apache-ssl> cat .\conf\httpd-ssl.conf
# MÃ³duls necessaris
LoadModule mpm_event_module modules/mod_mpm_event.so
LoadModule authz_core_module modules/mod_authz_core.so
LoadModule dir_module modules/mod_dir.so
LoadModule mime_module modules/mod_mime.so
LoadModule log_config_module modules/mod_log_config.so
LoadModule unixd_module modules/mod_unixd.so
LoadModule headers_module modules/mod_headers.so
LoadModule ssl_module modules/mod_ssl.so
LoadModule socache_shmcb_module modules/mod_socache_shmcb.so
LoadModule rewrite_module modules/mod_rewrite.so

ServerName localhost
# Sense Listen, Apache s'atura: "no listening sockets available"
Listen 80
Listen 443

User daemon
Group daemon

TypesConfig conf/mime.types
DirectoryIndex index.html

# ConfiguraciÃ³ de logging
ErrorLog "logs/error.log"
LogLevel warn
LogFormat "%h %l %u %t \"%r\" %>s %b \"%{Referer}i\" \"%{User-Agent}i\"" combined
CustomLog "logs/access.log" combined

# Cache SSL per millorar el rendiment
SSLSessionCache "shmcb:/usr/local/apache2/logs/ssl_scache(512000)"
SSLSessionCacheTimeout 300

# Virtual Host HTTP (port 80) - Redirigeix a HTTPS
```

```
</VirtualHost>

# Virtual Host HTTPS (port 443)
<VirtualHost *:443>
    ServerName localhost
    DocumentRoot "/usr/local/apache2/htdocs"

    # Activa SSL
    SSLEngine on

    # Rutes als certificats
    SSLCertificateFile "/usr/local/apache2/certs/server.crt"
    SSLCertificateKeyFile "/usr/local/apache2/certs/server.key"

    # Protocols SSL/TLS moderns (desactiva versions antigues insegures)
    # Seguint les recomanacions de seguretat 2025
    SSLProtocol all -SSLv3 -TLSv1 -TLSv1.1

    # Cipher suites segures (prioritza algoritmes moderns)
    SSLCipherSuite HIGH:!aNULL:!MD5:!RC4
    SSLHonorCipherOrder on

    # CapÃ§aleres de seguretat HTTP
    Header always set Strict-Transport-Security "max-age=31536000; includeSubDomains"
    Header always set X-Frame-Options "SAMEORIGIN"
    Header always set X-Content-Type-Options "nosniff"
    Header always set X-XSS-Protection "1; mode=block"

    <Directory "/usr/local/apache2/htdocs">
        Options -Indexes +FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>

    # Log d'errors HTTPS
ErrorLog /proc/self/fd/2

# Log estÃ ndard HTTPS
CustomLog /proc/self/fd/1 combined

# Log amb temps de resposta
CustomLog "logs/response-time.log" with_response_time

PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

Index.html amb certificat autosignat i redirecció HTTP a HTTPS, registra si la connexió és segura.

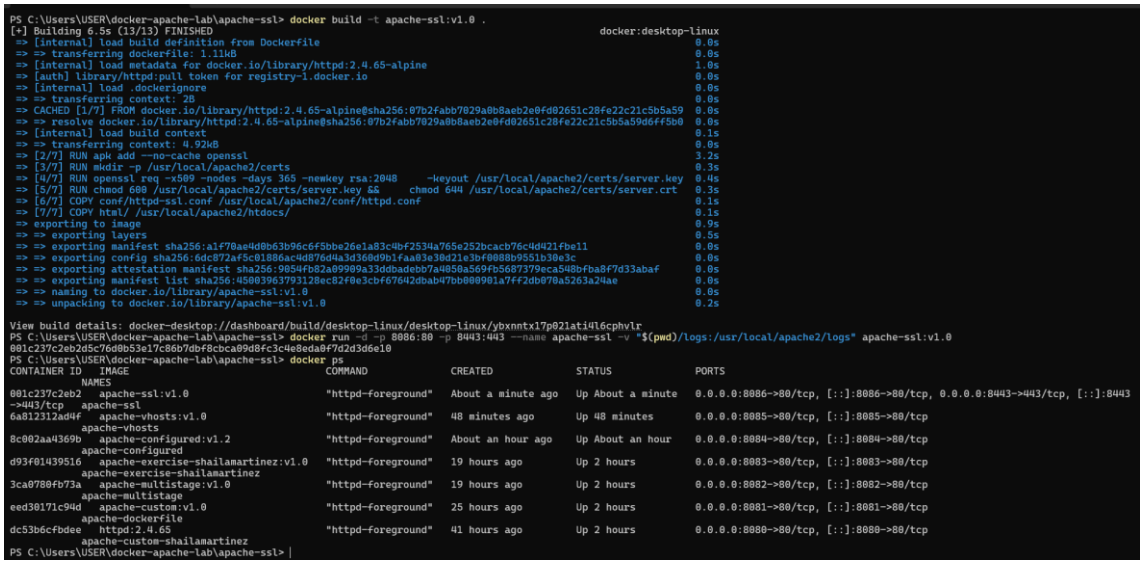
```
PS C:\Users\USER\docker-apache-lab\apache-ssl> cat .\html\index.html
<!DOCTYPE html>
<html lang="ca">
<head>
    <meta charset="UTF-8">
    <title>ConnexiÃ³ Segura HTTPS</title>
    <style>
        body {
            font-family: 'Arial', sans-serif;
            background: linear-gradient(135deg, #11998e 0%, #38ef7d 100%);
            color: white;
            padding: 50px;
            text-align: center;
        }
        .secure-badge {
            background: rgba(255, 255, 255, 0.2);
            backdrop-filter: blur(10px);
            padding: 30px;
            border-radius: 20px;
            display: inline-block;
            margin: 30px 0;
        }
        .lock { font-size: 5em; margin: 20px 0; }
        .info {
            background: rgba(0, 0, 0, 0.2);
            padding: 20px;
            border-radius: 10px;
            margin: 20px auto;
            max-width: 600px;
            text-align: left;
        }
    </style>
</head>
<body>
    <div class="secure-badge">
        <div class="lock"></div>
        <h1>ConnexiÃ³ Segura HTTPS</h1>
        <p>Apache 2.4.65 amb SSL/TLS activat</p>
    </div>

    <div class="info">
        <h3>InformaciÃ³ SSL/TLS:</h3>
        <ul>
            <li>Protocol: TLS 1.2 o superior</li>
            <li>Certificat: Auto-signat (nomÃ©s desenvolupament)</li>
            <li>HSTS: Activat (31536000 segons)</li>
            <li>RedirecciÃ³ HTTP â†' HTTPS: Activa</li>
        </ul>
    </div>

    <p><small>El navegador mostrarÃ  un avÃs perquÃ¨ el certificat Ã©s auto-signat.<br>
    En producciÃ³, utilitza certificats d'una CA reconeguda (Let's Encrypt, etc.)</small></p>

    <script>
        // Mostra informaciÃ³ sobre la connexiÃ³ (només funciona en HTTPS)
        if (location.protocol === 'https:') {
            console.log('ConnexiÃ³ segura HTTPS establerta');
        } else {
            console.log('Hauries de ser redirigit a HTTPS...');
        }
    </script>
</body>
</html>
PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

Creo la imatge amb Apache i un certificat SSL autofirmat. Després inicio el contenidor i configuro els ports: el 8086 per a HTTP i el 8443 per a HTTPS. Amb docker ps comprovo que els dos ports estan correctament configurats.

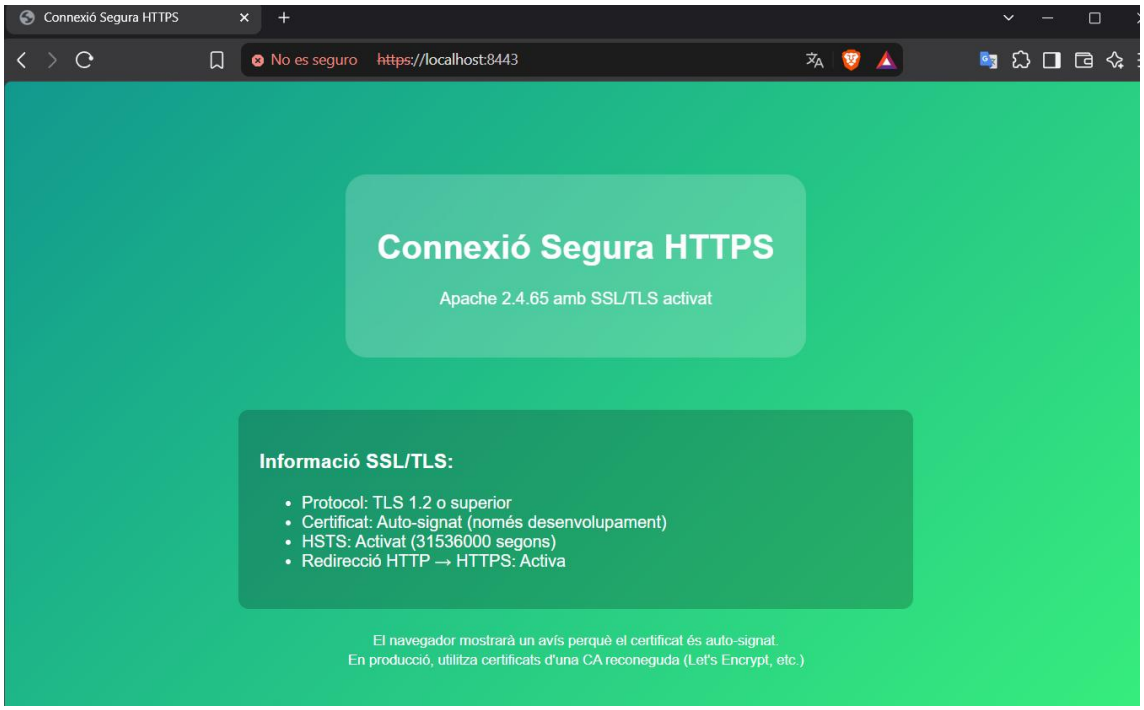


Quan accedeixo pel port 8086 (HTTP), Apache em redirigeix automàticament al port 8443 (HTTPS). Així la pàgina s'acaba carregant a través d'una connexió segura.

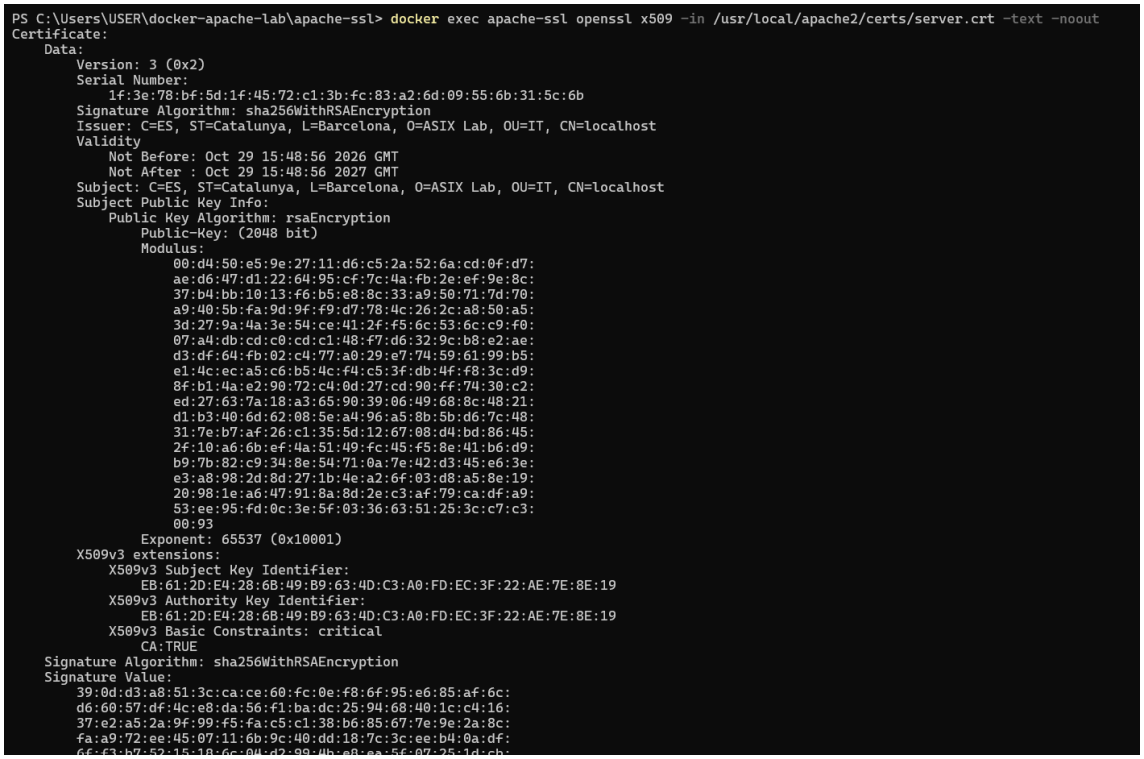
```
PS C:\Users\USER\docker-apache-lab\apache-ssl> curl.exe -I http://localhost:8086/
HTTP/1.1 301 Moved Permanently
Date: Sat, 07 Nov 2026 20:10:31 GMT
Server: Apache/2.4.65 (Unix) OpenSSL/3.5.9
Location: https://localhost:8443/
Content-Type: text/html; charset=iso-8859-1

PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

Localhost:8443 es veu la pàgina "Connexió Segura HTTPS" marca com a no segur (certificat autosignat).



Inspecció del certificat mostra: la versió, l'algoritme, emissor i subjecte són iguals (prova que és autosignat), validesa, clau pública...



Mod_rewrite avançat - Reescriptura d'URLs

Estructura apache-ssl i fitxer .htaccess: Configuro el fitxer .htaccess amb set regles per gestionar les URL, redirigir pàgines antigues, obligar a utilitzar HTTPS en zones concretes, bloquejar bots i fer redireccions segons l'adreça IP del client.

```
PS C:\Users\USER\docker-apache-lab\apache-ssl> tree /F
Listado de rutas de carpetas
El número de serie del volumen es 38C8-F55B
C:.
|   Dockerfile
|
+---certs
+---conf
|       httpd-ssl.conf
|
+---html
|       .htaccess
|       index.html
|
\---logs
        access.log
        error.log
        http-access.log
        http-error.log
        httpd.pid
        https-access.log
        https-error.log

PS C:\Users\USER\docker-apache-lab\apache-ssl> cat .\html\.htaccess
RewriteEngine On

# 1. RedirecciÃ³ de www a no-www (o viceversa)
RewriteCond %{HTTP_HOST} ^www\.(.*)$ [NC]
RewriteRule ^(.*)$ https://%1/$1 [R=301,L]

# 2. URLs amigables: /product/123 â†’ /product.php?id=123
RewriteRule ^product/([0-9]+)$ product.php?id=$1 [L,QSA]

# 3. Eliminar l'extensiÃ³ .php de les URLs
RewriteCond %{REQUEST_FILENAME} !-d
RewriteCond %{REQUEST_FILENAME}.php -f
RewriteRule ^(.+)$ $1.php [L]

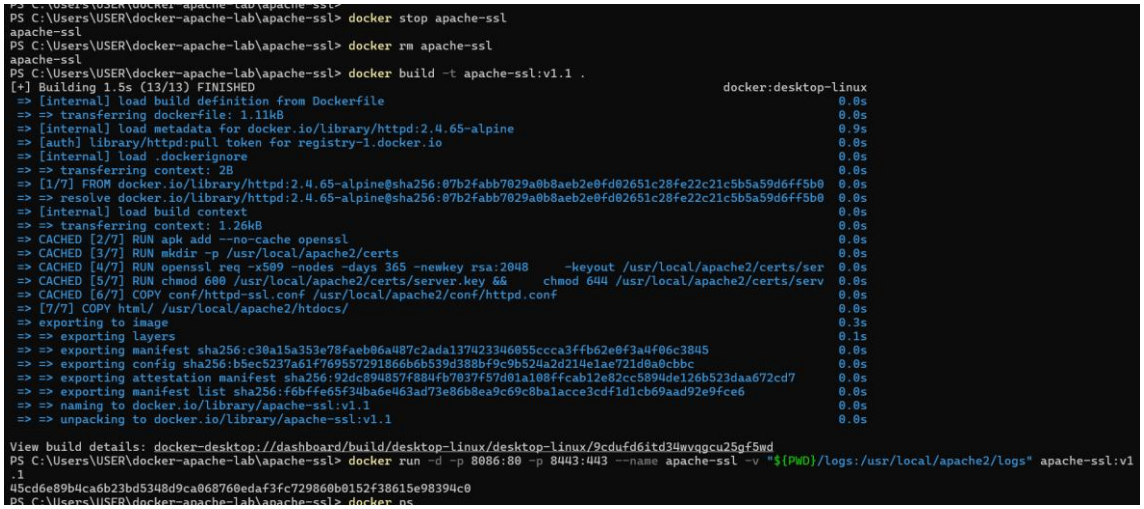
# 4. RedirecciÃ³ d'URLs antigues a noves
RewriteRule ^old-blog/(.*)$ /new-blog/$1 [R=301,L]

# 5. ForÃ§ar HTTPS per a certes pÃ gines
RewriteCond %{HTTPS} off
RewriteCond %{REQUEST_URI} ^/admin [OR]
RewriteCond %{REQUEST_URI} ^/login
RewriteRule ^(.*)$ https://%{HTTP_HOST}/$1 [R=301,L]

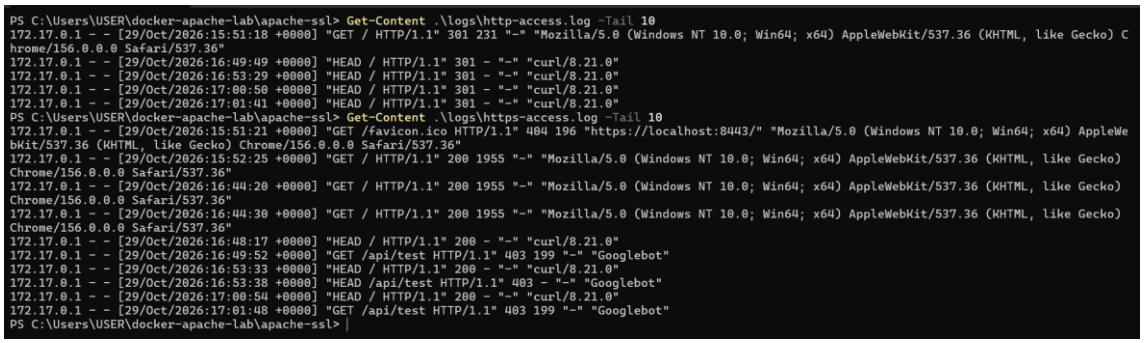
# 6. Bloquejar accÃ©s basat en User-Agent (anti-bots)
RewriteCond %{HTTP_USER_AGENT} ^.*(bot|spider|crawler).* [NC]
RewriteRule ^api/ - [F,L]

# 7. RedirecciÃ³ basada en la IP del client (geolocalitzaciÃ³ bÃ sica)
# Exemple: Si la IP comenÃ§a per 192.168, redirigeix a versiÃ³ local
RewriteCond %{REMOTE_ADDR} ^192\.168\.
RewriteRule ^$ /local-version.html [L]
PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

Faig docker stop i docker rm apache-ssl, construeixo apache-ssl:v1.1 i torno a executar.



Al fitxer http-access.log es veuen les peticions que es redirigeixen a HTTPS amb el codi 301. Al https-access.log es registren les peticions correctes amb codi 200. També comprovo que els bots, com Googlebot, reben un error 403 perquè estan bloquejats per la regla anti-bots. Les peticions normals de curl retornen 200 o 404.



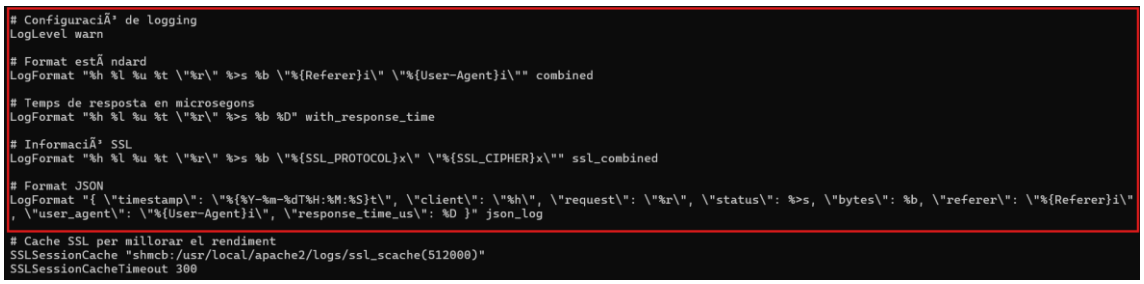
Configuració de logging personalitzat

A l'inici de httpd-ssl.conf s'afegeix mod_setenvif.so que és necessari per a les directives SetEnvIf del logging condicional.

```
PS C:\Users\USER\docker-apache-lab\apache-ssl> cat .\conf\httpd-ssl.conf
# MÃ²duls necessaris
LoadModule setenvif_module modules/mod_setenvif.so
LoadModule mpm_event_module modules/mod_mpm_event.so
LoadModule authz_core_module modules/mod_authz_core.so
LoadModule dir_module modules/mod_dir.so
LoadModule mime_module modules/mod_mime.so
LoadModule log_config_module modules/mod_log_config.so
LoadModule unixd_module modules/mod_unixd.so
LoadModule headers_module modules/mod_headers.so
LoadModule ssl_module modules/mod_ssl.so
LoadModule socache_shmcb_module modules/mod_socache_shmcb.so
LoadModule rewrite_module modules/mod_rewrite.so

ServerName localhost
# Sense Listen, Apache s'atura: "no listening sockets available"
Listen 80
Listen 443
```

Dins de httpd-ssl.conf defineixo 4 formats: combined (estàndard), with_response_time (microsegons), ssl_combined (afegeix ssl protocol i ssl cipher) i json_log.



```
# ConfiguraciÃ³ de logging
LogLevel warn

# Format estÃ ndard
LogFormat "%h %l %u %t \"%r\" %>s %b \"%{Referer}i\" \"%{User-Agent}i\"" combined

# Temps de resposta en microsegons
LogFormat "%h %l %u %t \"%r\" %>s %b %D" with_response_time

# InformaciÃ³ SSL
LogFormat "%h %l %u %t \"%r\" %>s %b \"%{SSL_PROTOCOL}x\" \"%{SSL_CIPHER}x\"" ssl_combined

# Format JSON
LogFormat "{ \"timestamp\": \"%{%Y-%m-%dT%H:%M:%S}t\", \"client\": \"%h\", \"request\": \"%r\", \"status\": %>s, \"bytes\": %b, \"referer\": \"%{Referer}i\"
, \"user_agent\": \"%{User-Agent}i\", \"response_time_us\": %D }" json_log

# Cache SSL per millorar el rendiment
SSLSessionCache "shmcb:/usr/local/apache2/logs/ssl_scache(512000)"
SSLSessionCacheTimeout 300
```

Dins del VirtualHost:443 configuro diferents logs per registrar els temps de resposta, connexions SSL, peticions JSON, els errors i les imatges. També faig que les peticions a /health no apareguin al log general. Comprovo que la configuració no té errors.

```
    # Log d'errors HTTPS
ErrorLog /proc/self/fd/2

# Log estÃ ndard HTTPS
CustomLog /proc/self/fd/1 combined

# Log amb temps de resposta
CustomLog "logs/response-time.log" with_response_time

# Log amb informaciÃ³ SSL
CustomLog "logs/ssl-access.log" ssl_combined

# Log en format JSON
CustomLog "logs/json-access.log" json_log

# Registrar nomÃ©s errors 4xx i 5xx
CustomLog "logs/errors-only.log" combined "expr=%{REQUEST_STATUS} >= 400"

# Logs d'imatges
SetEnvIf Request_URI "\.jpg$" image-request
SetEnvIf Request_URI "\.png$" image-request
SetEnvIf Request_URI "\.gif$" image-request
CustomLog "logs/images.log" combined env=image-request

# Excloure healthchecks del log d'accÃ©s
SetEnvIf Request_URI "^/health$" dontlog
CustomLog "logs/access.log" combined env=!dontlog
</VirtualHost>
PS C:\Users\USER\docker-apache-lab\apache-ssl> docker exec apache-ssl httpd -t
Syntax OK
PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

Seguidament faig docker stop apache-ssl, construeixo v1.2 i torno a validar.

```
PS C:\Users\USER\docker-apache-lab\apache-ssl> docker stop apache-ssl
apache-ssl
PS C:\Users\USER\docker-apache-lab\apache-ssl> docker rm apache-ssl
apache-ssl
PS C:\Users\USER\docker-apache-lab\apache-ssl> docker build -t apache-ssl:v1.2 .
[+] Building 1.7s (13/13) FINISHED                                                
```

El fitxer images.log només registra les peticions d'imatges (.jpg, .png i .gif). Encara que prova.jpg no existeixi i retorni un error 404, la petició queda registrada igualment perquè es té en compte la URL.

```
PS C:\Users\USER\docker-apache-lab\apache-ssl> curl.exe -k -I https://localhost:8443/prova.jpg
HTTP/1.1 404 Not Found
Date: Sat, 07 Nov 2026 20:26:56 GMT
Server: Apache/2.4.65 (Unix) OpenSSL/3.5.9
Strict-Transport-Security: max-age=31536000; includeSubDomains
X-Frame-Options: SAMEORIGIN
X-Content-Type-Options: nosniff
X-XSS-Protection: 1; mode=block
Content-Type: text/html; charset=iso-8859-1

PS C:\Users\USER\docker-apache-lab\apache-ssl> cat .\logs\images.log -Tail 5
172.17.0.1 - - [07/Nov/2026:20:26:56 +0000] "HEAD /prova.jpg HTTP/1.1" 404 - "-" "curl/8.21.0"
PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

Puc consultar els logs amb docker logs apache-ssl sense entrar al contenidor. Les peticions HTTPS apareixen registrades amb el format combined. En canvi, les peticions del virtual host secure-site es guarden en els seus propis fitxers de log.

```
PS C:\Users\USER\docker-apache-lab\apache-ssl> curl.exe -k -I https://localhost:8443/
HTTP/1.1 200 OK
Date: Sat, 07 Nov 2026 20:24:30 GMT
Server: Apache/2.4.65 (Unix) OpenSSL/3.5.9
Strict-Transport-Security: max-age=31536000; includeSubDomains
X-Frame-Options: SAMEORIGIN
X-Content-Type-Options: nosniff
X-XSS-Protection: 1; mode=block
Last-Modified: Thu, 29 Oct 2026 15:44:59 GMT
ETag: "7a3-65efc8fe234c0"
Accept-Ranges: bytes
Content-Length: 1955
Content-Type: text/html

PS C:\Users\USER\docker-apache-lab\apache-ssl> docker logs apache-ssl --tail 10
172.17.0.1 - - [07/Nov/2026:20:14:53 +0000] "HEAD /api/test HTTP/1.1" 404 - "-" "Googlebot"
172.17.0.1 - - [07/Nov/2026:20:24:30 +0000] "HEAD / HTTP/1.1" 200 - "-" "curl/8.21.0"
PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

Repte 4

Estructura final de apache-ssl: nova carpeta secure-site amb .htaccess, about.html, index.html i private\secret.txt i a logs tots els fitxers nous.

```
PS C:\Users\USER\docker-apache-lab\apache-ssl> tree /f
Listado de rutas de carpetas
El número de serie del volumen es 38C8-F55B
C:.
|   Dockerfile
|
+---certs
+---conf
|       httpd-ssl.conf
|
+---html
|   |   index.html
|   |
|   \---secure-site
|       |   .htaccess
|       |   about.html
|       |   index.html
|       |
|       \---private
|               secret.txt
|
\---logs
        access.log
        error.log
        errors-only.log
        http-access.log
        http-error.log
        httpd.pid
        https-access.log
        https-error.log
        images.log
        json-access.log
        response-time.log
        secure-site-access.log
        secure-site-error.log
        ssl-access.log

PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

El fitxer .htaccess bloqueja l'accés a /private amb un error 403, permet entrar a les pàgines sense escriure l'extensió .html i afegeix la capçalera X-Secure-Site: true a totes les respostes.

```
PS C:\Users\USER\docker-apache-lab\apache-ssl> cat .\html\secure-site\.htaccess
RewriteEngine On

# Bloquejar el directori private
RewriteRule ^private(/|$) - [F,L]

# Eliminar .html de les URLs
RewriteCond %{REQUEST_FILENAME} !-d
RewriteCond %{REQUEST_FILENAME}.html -f
RewriteRule ^(.+)$ $1.html [L]

# Capçalera de seguretat
Header always set X-Secure-Site "true"
PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

Configuro el Virtual Host perquè totes les connexions HTTP es redirigeixin a HTTPS pel port 8443. Defineixo el directori propi de la web, activo SSL i permeto l'ús de .htaccess. Configuro dos fitxers de log separats per guardar els errors i els accessos.

```
PS C:\Users\USER\docker-apache-lab\apache-ssl> cat .\conf\httpd-ssl.conf -Tail 40
CustomLog "logs/json-access.log" json_log

# Registrar nomÃ©s errors 4xx i 5xx
CustomLog "logs/errors-only.log" combined "expr=%{REQUEST_STATUS} >= 400"

# Logs d'imatges
SetEnvIf Request_URI "\.jpg$" image-request
SetEnvIf Request_URI "\.png$" image-request
SetEnvIf Request_URI "\.gif$" image-request
CustomLog "logs/images.log" combined env=image-request

# Excloure healthchecks del log d'accÃ©s
SetEnvIf Request_URI "^/health$" dontlog
CustomLog "logs/access.log" combined env=!dontlog
</VirtualHost>

# Virtual Host HTTP: redirecciÃ³ a HTTPS
<VirtualHost *:80>
    ServerName secure-site-shailamartinez.local
    Redirect permanent / https://secure-site-shailamartinez.local:8443/
</VirtualHost>

# Virtual Host HTTPS
<VirtualHost *:443>
    ServerName secure-site-shailamartinez.local
    DocumentRoot "/usr/local/apache2/htdocs/secure-site"

    SSLEngine on
    SSLCertificateFile "/usr/local/apache2/certs/server.crt"
    SSLCertificateKeyFile "/usr/local/apache2/certs/server.key"

    <Directory "/usr/local/apache2/htdocs/secure-site">
        Options -Indexes +FollowSymLinks
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog "logs/secure-site-error.log"
    CustomLog "logs/secure-site-access.log" combined
</VirtualHost>
PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

Els logs d'aquest Virtual Host són independents de la resta de pàgines web.
Comprovo que la configuració és correcta, retorna Syntax OK.

```
PS C:\Users\USER\docker-apache-lab\apache-ssl> Select-String -Path .\conf\httpd-ssl.conf -Pattern 'LogFormat|ErrorLog|Cu
stomLog'

conf\httpd-ssl.conf:30:LogFormat "%h %l %u %t \"%r\" %>s %b \"%{Referer}i\" \"%{User-Agent}i\"" combined
conf\httpd-ssl.conf:33:LogFormat "%h %l %u %t \"%r\" %>s %b %D" with_response_time
conf\httpd-ssl.conf:36:LogFormat "%h %l %u %t \"%r\" %>s %b \"%{SSL_PROTOCOL}x\" \"%{SSL_CIPHER}x\"" ssl_combined
conf\httpd-ssl.conf:39:LogFormat "{ \"timestamp\": \"%{%Y-%m-%dT%H:%M:%S}t\", \"client\": \"%h\", \"request\": \"%r\",
\"status\": %>s, \"bytes\": %b, \"referer\": \"%{Referer}i\", \"user_agent\": \"%{User-Agent}i\",
\"response_time_us\": %D }" json_log
conf\httpd-ssl.conf:63:    ErrorLog "logs/http-error.log"
conf\httpd-ssl.conf:64:    CustomLog "logs/http-access.log" combined
conf\httpd-ssl.conf:100:ErrorLog /proc/self/fd/2
conf\httpd-ssl.conf:103:CustomLog /proc/self/fd/1 combined
conf\httpd-ssl.conf:106:CustomLog "logs/response-time.log" with_response_time
conf\httpd-ssl.conf:109:CustomLog "logs/ssl-access.log" ssl_combined
conf\httpd-ssl.conf:112:CustomLog "logs/json-access.log" json_log
conf\httpd-ssl.conf:115:CustomLog "logs/errors-only.log" combined "expr=%{REQUEST_STATUS} >= 400"
conf\httpd-ssl.conf:121:CustomLog "logs/images.log" combined env=image-request
conf\httpd-ssl.conf:125:CustomLog "logs/access.log" combined env=!dontlog
conf\httpd-ssl.conf:149:    ErrorLog "logs/secure-site-error.log"
conf\httpd-ssl.conf:150:    CustomLog "logs/secure-site-access.log" combined


PS C:\Users\USER\docker-apache-lab\apache-ssl> docker exec apache-ssl httpd -t
Syntax OK
PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

Comprovo que la pàgina principal i /about retornen 200, /private queda bloquejat amb un 403 i les connexions HTTP es redirigeixen automàticament a HTTPS amb un 301. Verifico que s'afegeix correctament la capçalera X-Secure-Site: true.

```
PS C:\Users\USER\docker-apache-lab\apache-ssl>
PS C:\Users\USER\docker-apache-lab\apache-ssl> curl.exe -k -I --resolve secure-site-shailamartinez.local:8443:127.0.0.1 https://secure-site-shailamartinez.l
ocal:8443/
HTTP/1.1 200 OK
Date: Thu, 29 Oct 2026 20:27:50 GMT
Server: Apache/2.4.65 (Unix) OpenSSL/3.5.9
X-Secure-Site: true
Last-Modified: Thu, 29 Oct 2026 17:28:46 GMT
ETag: "d1-65efe030aaf80"
Accept-Ranges: bytes
Content-Length: 209
Content-Type: text/html

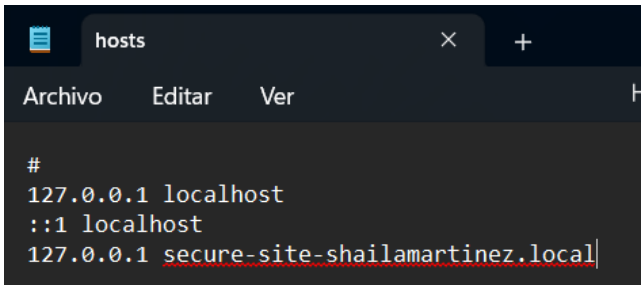
PS C:\Users\USER\docker-apache-lab\apache-ssl> curl.exe -k -I --resolve secure-site-shailamartinez.local:8443:127.0.0.1 https://secure-site-shailamartinez.l
ocal:8443/about
HTTP/1.1 200 OK
Date: Thu, 29 Oct 2026 20:27:54 GMT
Server: Apache/2.4.65 (Unix) OpenSSL/3.5.9
X-Secure-Site: true
Last-Modified: Thu, 29 Oct 2026 17:28:22 GMT
ETag: "8b-65efe019c7980"
Accept-Ranges: bytes
Content-Length: 139
Content-Type: text/html

PS C:\Users\USER\docker-apache-lab\apache-ssl> curl.exe -k -I --resolve secure-site-shailamartinez.local:8443:127.0.0.1 https://secure-site-shailamartinez.l
ocal:8443/private/secret.txt
HTTP/1.1 403 Forbidden
Date: Thu, 29 Oct 2026 20:27:58 GMT
Server: Apache/2.4.65 (Unix) OpenSSL/3.5.9
X-Secure-Site: true
Content-Type: text/html; charset=iso-8859-1

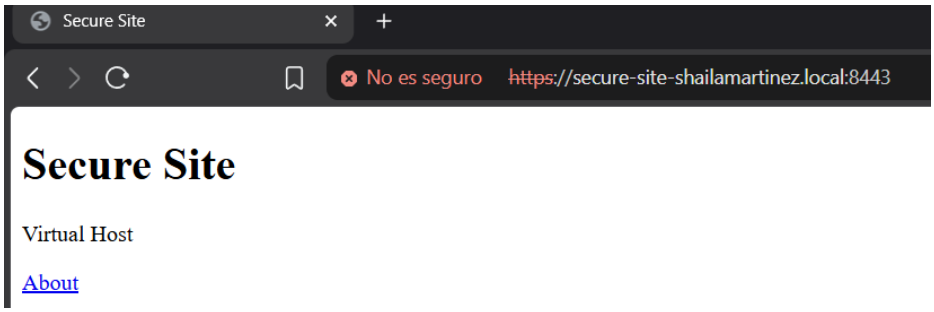
PS C:\Users\USER\docker-apache-lab\apache-ssl> curl.exe -I -H "Host: secure-site-shailamartinez.local" http://localhost:8086/
HTTP/1.1 301 Moved Permanently
Date: Thu, 29 Oct 2026 20:28:00 GMT
Server: Apache/2.4.65 (Unix) OpenSSL/3.5.9
Location: https://secure-site-shailamartinez.local:8443/
Content-Type: text/html; charset=iso-8859-1

PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

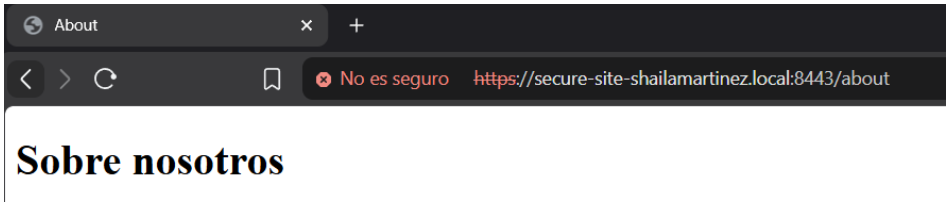
Per resoldre el domini en local s'afegeix 127.0.0.1 secure-site-shailamartinez.local al fitxer hosts.



Al navegador es carreguen https://secure-site-shailamartinez.local:8443 i /about sense l'extensió .html



About:



Apache també estava llegint el .htaccess del directori principal (htdocs), cosa que provocava un error. Per solucionar-ho el vaig treure i vaig deixar només el de secure-site.

Paro el contenidor, netejo els logs i l'arrenco.

```
PS C:\Users\USER\docker-apache-lab\apache-ssl> docker stop apache-ssl
apache-ssl
PS C:\Users\USER\docker-apache-lab\apache-ssl> Remove-Item .\logs\secure-site-*.log
PS C:\Users\USER\docker-apache-lab\apache-ssl> docker start apache-ssl
apache-ssl
PS C:\Users\USER\docker-apache-lab\apache-ssl> curl.exe -k -I --resolve secure-site-shailamartinez.local:8443:127.0.0.1 https://secure-site-shailamartinez.local:8443/
HTTP/1.1 200 OK
Date: Sat, 07 Nov 2026 22:13:44 GMT
Server: Apache/2.4.65 (Unix) OpenSSL/3.5.9
X-Secure-Site: true
Last-Modified: Thu, 29 Oct 2026 17:28:46 GMT
ETag: "d1-65efe030aaf80"
Accept-Ranges: bytes
Content-Length: 209
Content-Type: text/html

PS C:\Users\USER\docker-apache-lab\apache-ssl> curl.exe -k -I --resolve secure-site-shailamartinez.local:8443:127.0.0.1 https://secure-site-shailamartinez.local:8443/about
HTTP/1.1 200 OK
Date: Sat, 07 Nov 2026 22:13:49 GMT
Server: Apache/2.4.65 (Unix) OpenSSL/3.5.9
X-Secure-Site: true
Last-Modified: Thu, 29 Oct 2026 17:28:22 GMT
ETag: "8b-65efe019c7980"
Accept-Ranges: bytes
Content-Length: 139
Content-Type: text/html

PS C:\Users\USER\docker-apache-lab\apache-ssl> curl.exe -k -I --resolve secure-site-shailamartinez.local:8443:127.0.0.1 https://secure-site-shailamartinez.local:8443/private/secret.txt
HTTP/1.1 403 Forbidden
Date: Sat, 07 Nov 2026 22:13:53 GMT
Server: Apache/2.4.65 (Unix) OpenSSL/3.5.9
X-Secure-Site: true
Content-Type: text/html; charset=iso-8859-1

PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

El secure-site-access.log registra només les peticions d'aquest Virtual Host.

```
PS C:\Users\USER\docker-apache-lab\apache-ssl> cat .\logs\secure-site-access.log
172.17.0.1 - - [07/Nov/2026:22:13:44 +0000] "HEAD / HTTP/1.1" 200 - "-" "curl/8.21.0"
172.17.0.1 - - [07/Nov/2026:22:13:49 +0000] "HEAD /about HTTP/1.1" 200 - "-" "curl/8.21.0"
172.17.0.1 - - [07/Nov/2026:22:13:53 +0000] "HEAD /private/secret.txt HTTP/1.1" 403 - "-" "curl/8.21.0"
PS C:\Users\USER\docker-apache-lab\apache-ssl> cat .\logs\secure-site-error.log
[Sat Nov 07 22:13:27.533838 2026] [ssl:warn] [pid 1:tid 1] AH01906: secure-site-shailamartinez.local:443:0 server certificate is a CA certificate (BasicConstraints: CA == TRUE !?)
[Sat Nov 07 22:13:27.534362 2026] [ssl:warn] [pid 1:tid 1] AH01909: secure-site-shailamartinez.local:443:0 server certificate does NOT include an ID which matches the server name
[Sat Nov 07 22:13:27.560825 2026] [ssl:warn] [pid 1:tid 1] AH01906: secure-site-shailamartinez.local:443:0 server certificate is a CA certificate (BasicConstraints: CA == TRUE !?)
[Sat Nov 07 22:13:27.561902 2026] [ssl:warn] [pid 1:tid 1] AH01909: secure-site-shailamartinez.local:443:0 server certificate does NOT include an ID which matches the server name
PS C:\Users\USER\docker-apache-lab\apache-ssl>
```

6. Part 5: Docker Compose - Orquestració de contenidors

Exemple pràctic: Apache + MySQL + phpMyAdmin

Estructura docker-compose-stack i Dockerfile que instal·la i activa l'extensió mysqli per poder connectar amb Mysql, activa el mòdul rewrite, copia el contingut web a /var/www/html.

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> tree /f
Listado de rutas de carpetas
El número de serie del volumen es 38C8-F55B
C:.
|   docker-compose.yml
|
+---apache
|   |   Dockerfile
|   |
|   +---conf
|   |       httpd.conf
|   |
|   \---html
|           index.php
|
+---logs
\---mysql-data
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> cat .\apache\Dockerfile
FROM php:8.2-apache

LABEL maintainer="s.martinez6@sapalomera.cat"
LABEL description="Apache amb PHP i extensiÃ³ MySQL"

# InstalÂ·la l'extensiÃ³ mysqli per connectar a MySQL
RUN docker-php-ext-install mysqli && docker-php-ext-enable mysqli

# Activa mod_rewrite
RUN a2enmod rewrite

# Copia el contingut web
COPY html/ /var/www/html/

# Exposa el port 80
EXPOSE 80
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
```

Pàgina php que connecta amb Mysql. Crea la taula visits si no existeix, hi insereix cada visita, mostra la versió de Mysql...

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> cat .\apache\html\index.php
<!DOCTYPE html>
<html lang="ca">
<head>
    <meta charset="UTF-8">
    <title>Stack LAMP amb Docker Compose</title>
    <style>
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            color: white;
            padding: 40px;
            margin: 0;
        }
        .container {
            max-width: 800px;
            margin: 0 auto;
            background: rgba(255, 255, 255, 0.1);
            backdrop-filter: blur(10px);
            padding: 40px;
            border-radius: 20px;
        }
        .status {
            padding: 20px;
            margin: 20px 0;
            border-radius: 10px;
            background: rgba(255, 255, 255, 0.2);
        }
        .success { border-left: 5px solid #2ecc71; }
        .error { border-left: 5px solid #e74c3c; }
        code {
            background: rgba(0, 0, 0, 0.3);
            padding: 2px 8px;
            border-radius: 3px;
        }
        table {
            width: 100%;
```

Fitxer docker-compose que defineix els 3 serveis: apache (port 8090 amb bind mounts per al codi i els logs i dependent de Mysql), mysql (volum amb nom mysql-

data i healthcheck) i phpmyadmin (port 8091). El contingut complet ja amb variables d'entorn, es mostra a l'apartat Variables d'entorn i fitxers .env

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> cat .\docker-compose.yml
# DefiniciÃ³ dels serveis (contenidors)
services:

  # Servei Apache amb PHP
  apache:
    # Construeix la imatge des del Dockerfile
    build:
      context: ./apache
      dockerfile: Dockerfile

    # Nom del contenidor
    container_name: lamp-apache

    # Reinicia automÃ ticament si el contenidor falla
    restart: unless-stopped

    # Mapatge de ports: host:container
    ports:
      - "8090:80"

    # Volums: persistÃ¨ncia i desenvolupament
    volumes:
      - ./apache/html:/var/www/html  # Codi PHP (bind mount)
      - ./logs:/var/log/apache2      # Logs d'Apache

    # Variables d'entorn
    environment:
      - TZ=Europe/Madrid

    # DepÃ¨n del servei MySQL: no arrenca fins que el healthcheck diu que estÃ  llest
    depends_on:
      mysql:
        condition: service_healthy
```

```
      resources:
        limits:
          cpus: '1.0'
          memory: 512M

  # Servei MySQL
  mysql:
    # Imatge oficial de MySQL (versiÃ³ especÃfica, no "latest")
    image: mysql:8.0.35

    container_name: lamp-mysql
    restart: unless-stopped

    # Variables d'entorn per configurar MySQL
    environment:
      - MYSQL_ROOT_PASSWORD=${MYSQL_ROOT_PASSWORD}
      - MYSQL_DATABASE=${MYSQL_DATABASE}
      - MYSQL_USER=${MYSQL_USER}
      - MYSQL_PASSWORD=${MYSQL_PASSWORD}
      - TZ=${TZ}

    # Volum amb nom per persistir dades de MySQL
    volumes:
      - mysql-data:/var/lib/mysql

    # Exposa el port MySQL (opcional, nomÃ©s si vols accedir des de l'host)
    ports:
      - "${MYSQL_PORT}:3306"

    networks:
      - lamp-network

    # Healthcheck per verificar que MySQL estÃ  llest
    healthcheck:
      test: ["CMD", "mysqladmin", "ping", "-h", "localhost", "-uroot", "-p${MYSQL_ROOT_PASSWORD}"]
      interval: 10s
      timeout: 5s
      retries: 3
      start_period: 30s

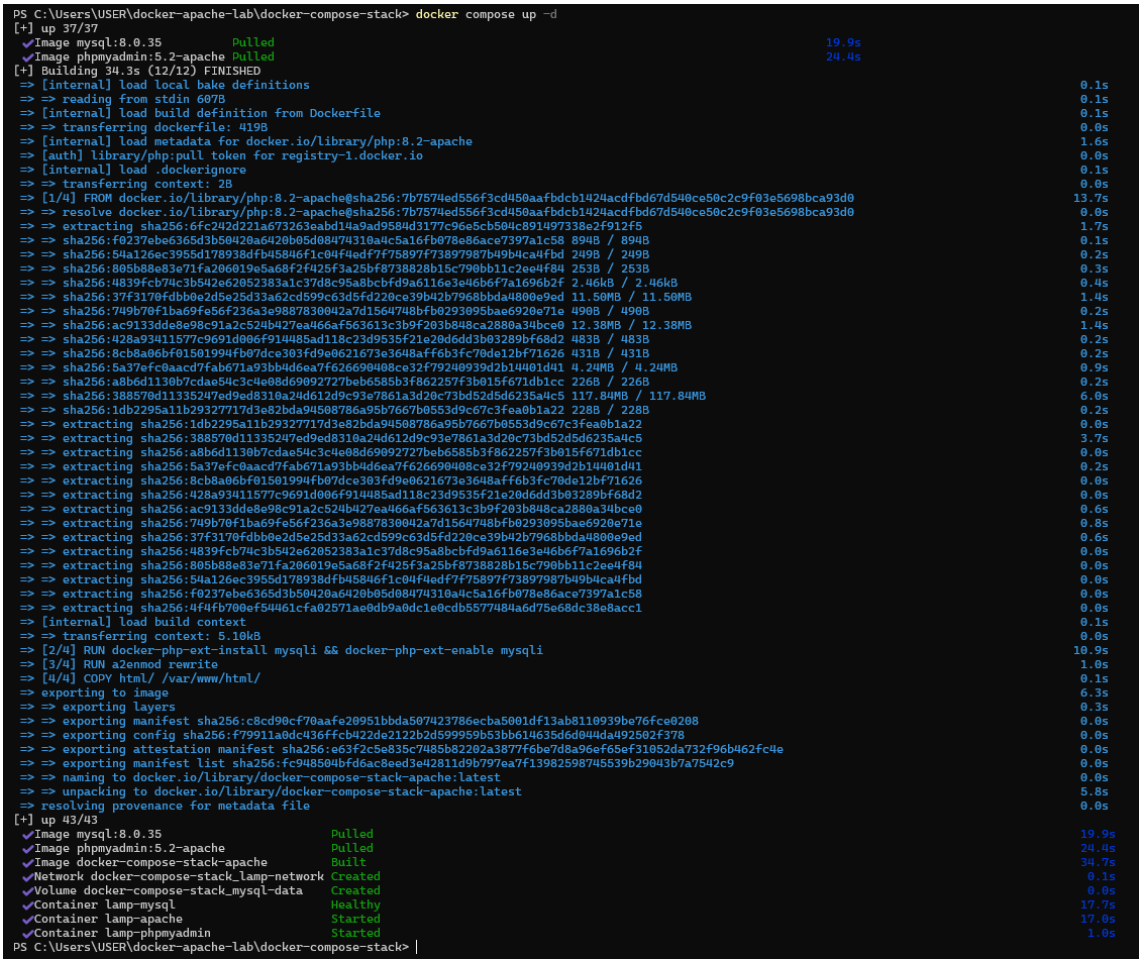
  # Servei phpMyAdmin (interfÃcie web per gestionar MySQL)
  phpmyadmin:
    image: phpmyadmin:5.2-apache
    container_name: lamp-phpmyadmin
    restart: unless-stopped

    ports:
      - "${PHPMYADMIN_PORT}:80"   # el 8081 ja l'ocupa el contenidor apache-dockerfile de la Part 2

    environment:
      - PMA_HOST=mysql
      - PMA_PORT=3306
      - PMA_USER=root
      - PMA_PASSWORD=${MYSQL_ROOT_PASSWORD}
      - UPLOAD_LIMIT=50M
```

Executar l'stack amb Docker Compose

Docker descarrega les imatges:



Amb docker compose logs es pot veure els registres dels tres serveis. Mysql mostra que s'ha creat la base de dades i l'usuari correctament. Apache i phpmyAdmin mostren un avís per no tenir configurat el ServerName, però no afecta el funcionament.

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker compose logs
lamp-phpmyadmin  | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.18.0.3. Set the 'ServerName' directive
 globally to suppress this message
lamp-mysql       | 2026-10-30 13:21:02+01:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.35-1.el8 started.
lamp-mysql       | 2026-10-30 13:21:03+01:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
lamp-mysql       | 2026-10-30 13:21:03+01:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.35-1.el8 started.
lamp-mysql       | 2026-10-30 13:21:03+01:00 [Note] [Entrypoint]: Initializing database files
lamp-mysql       | 2026-10-30T12:21:03.393910Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future
 release. Please use SET GLOBAL host_cache_size=0 instead.
lamp-mysql       | 2026-10-30T12:21:03.394078Z 0 [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 8.0.35) initializing of server in progress as proces
s 80
lamp-mysql       | 2026-10-30T12:21:03.406126Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
lamp-mysql       | 2026-10-30T12:21:04.360818Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
lamp-mysql       | 2026-10-30T12:21:09.049806Z 6 [Warning] [MY-010453] [Server] root@localhost is created with an empty password ! Please consider switching
 off the --initialize-insecure option.
lamp-mysql       | 2026-10-30 13:21:14+01:00 [Note] [Entrypoint]: Database files initialized
lamp-mysql       | 2026-10-30 13:21:14+01:00 [Note] [Entrypoint]: Starting temporary server
lamp-mysql       | 2026-10-30T12:21:14.315533Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future
 release. Please use SET GLOBAL host_cache_size=0 instead.
lamp-mysql       | 2026-10-30T12:21:14.317690Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.35) starting as process 135
lamp-mysql       | 2026-10-30T12:21:14.337772Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
lamp-mysql       | 2026-10-30T12:21:14.520806Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
lamp-mysql       | 2026-10-30T12:21:14.845050Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
lamp-mysql       | 2026-10-30T12:21:14.845099Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now s
upported for this channel.
lamp-mysql       | 2026-10-30T12:21:14.849849Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the pat
h is accessible to all OS users. Consider choosing a different directory.
lamp-mysql       | 2026-10-30T12:21:14.868705Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Socket: /var/run/mysqld/mysqlx.sock
lamp-mysql       | 2026-10-30T12:21:14.868779Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.35'  socket: '/var/run
/mysqld/mysqld.sock'  port: 0  MySQL Community Server - GPL.
lamp-phpmyadmin  | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.18.0.3. Set the 'ServerName' directive
 globally to suppress this message
lamp-phpmyadmin  | [Fri Oct 30 12:21:03.325733 2026] [mpm_prefork:notice] [pid 1:tid 1] AH00163: Apache/2.4.69 (Debian) PHP/8.3.36 configured -- resuming no
rmal operations
lamp-mysql       | 2026-10-30 13:21:14+01:00 [Note] [Entrypoint]: Temporary server started.
lamp-mysql       | '/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
lamp-mysql       | Warning: Unable to load '/usr/share/zoneinfo/iso3166.tab' as time zone. Skipping it.
lamp-mysql       | Warning: Unable to load '/usr/share/zoneinfo/leap-seconds.list' as time zone. Skipping it.
lamp-mysql       | Warning: Unable to load '/usr/share/zoneinfo/leapseconds' as time zone. Skipping it.
lamp-mysql       | Warning: Unable to load '/usr/share/zoneinfo/tzdata.zi' as time zone. Skipping it.
```

Logs:

```
lamp-mysql       | Warning: Unable to load '/usr/share/zoneinfo/tzdata.zi' as time zone. Skipping it.
lamp-mysql       | Warning: Unable to load '/usr/share/zoneinfo/zone.tab' as time zone. Skipping it.
lamp-apache      | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.18.0.4. Set the 'ServerName' directive
 globally to suppress this message
lamp-phpmyadmin  | [Fri Oct 30 12:21:03.325818 2026] [core:notice] [pid 1:tid 1] AH00094: Command line: 'apache2 -D FOREGROUND'
lamp-mysql       | Warning: Unable to load '/usr/share/zoneinfo/zone1970.tab' as time zone. Skipping it.
lamp-mysql       | 2026-10-30 13:21:21+01:00 [Note] [Entrypoint]: Creating database testdb
lamp-mysql       | 2026-10-30 13:21:21+01:00 [Note] [Entrypoint]: Creating user appuser
lamp-mysql       | 2026-10-30 13:21:21+01:00 [Note] [Entrypoint]: Giving user appuser access to schema testdb
lamp-mysql       |
lamp-mysql       | 2026-10-30 13:21:21+01:00 [Note] [Entrypoint]: Stopping temporary server
lamp-mysql       | 2026-10-30T12:21:21.151899Z 14 [System] [MY-013172] [Server] Received SHUTDOWN from user root. Shutting down mysqld (Version: 8.0.35).
lamp-mysql       | 2026-10-30T12:21:23.660175Z 0 [System] [MY-010910] [Server] /usr/sbin/mysqld: Shutdown complete (mysqld 8.0.35)  MySQL Community Server -
 GPL.
lamp-mysql       | 2026-10-30 13:21:24+01:00 [Note] [Entrypoint]: Temporary server stopped
lamp-mysql       |
lamp-mysql       | 2026-10-30 13:21:24+01:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.
lamp-mysql       |
lamp-mysql       | 2026-10-30T12:21:24.408888Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future
 release. Please use SET GLOBAL host_cache_size=0 instead.
lamp-mysql       | 2026-10-30T12:21:24.410244Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.35) starting as process 1
lamp-mysql       | 2026-10-30T12:21:24.418864Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
lamp-mysql       | 2026-10-30T12:21:24.575299Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
lamp-mysql       | 2026-10-30T12:21:24.874868Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
lamp-mysql       | 2026-10-30T12:21:24.874928Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now s
upported for this channel.
lamp-mysql       | 2026-10-30T12:21:24.880180Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the pat
h is accessible to all OS users. Consider choosing a different directory.
lamp-mysql       | 2026-10-30T12:21:24.905777Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/
run/mysqld/mysqlx.sock
lamp-mysql       | 2026-10-30T12:21:24.905967Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.35'  socket: '/var/run
/mysqld/mysqld.sock'  port: 3306  MySQL Community Server - GPL.
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
```

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker compose logs
lamp-phpmyadmin  | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.18
.0.3. Set the 'ServerName' directive globally to suppress this message
lamp-mysql       | 2026-10-30 13:21:02+01:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.35-1.el8 starte
d.
lamp-mysql       | 2026-10-30 13:21:03+01:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
lamp-mysql       | 2026-10-30 13:21:03+01:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.35-1.el8 starte
d.
lamp-mysql       | 2026-10-30 13:21:03+01:00 [Note] [Entrypoint]: Initializing database files
lamp-mysql       | 2026-10-30T12:21:03.393910Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprec
ated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
lamp-mysql       | 2026-10-30T12:21:03.394078Z 0 [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 8.0.35) initiali
zing of server in progress as process 80
lamp-mysql       | 2026-10-30T12:21:03.406126Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
lamp-mysql       | 2026-10-30T12:21:04.360818Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
lamp-mysql       | 2026-10-30T12:21:09.049806Z 6 [Warning] [MY-010453] [Server] root@localhost is created with an empty
password ! Please consider switching off the --initialize-insecure option.
lamp-mysql       | 2026-10-30 13:21:14+01:00 [Note] [Entrypoint]: Database files initialized
lamp-mysql       | 2026-10-30 13:21:14+01:00 [Note] [Entrypoint]: Starting temporary server
lamp-mysql       | 2026-10-30T12:21:14.315533Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprec
ated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
lamp-mysql       | 2026-10-30T12:21:14.317690Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.35) starting
 as process 135
lamp-mysql       | 2026-10-30T12:21:14.337772Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
lamp-mysql       | 2026-10-30T12:21:14.520806Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
lamp-mysql       | 2026-10-30T12:21:14.845050Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
lamp-mysql       | 2026-10-30T12:21:14.845099Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support
TLS. Encrypted connections are now supported for this channel.
```

Mostra només els logs del servei Apache.

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker compose logs -f apache
lamp-apache  | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.18.0.4
. Set the 'ServerName' directive globally to suppress this message
```

Els tres contenidors en Up.

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker compose ps
NAME              IMAGE                          COMMAND                  SERVICE      CREATED         STATUS
      PORTS
lamp-apache       docker-compose-stack-apache    "docker-php-entrypoi…"   apache       5 minutes ago   Up 4 minutes
      0.0.0.0:8090->80/tcp, [::]:8090->80/tcp
lamp-mysql        mysql:8.0.35                   "docker-entrypoint.s…"   mysql        5 minutes ago   Up 5 minutes (heal
thy)   0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp
lamp-phpmyadmin   phpmyadmin:5.2-apache          "/docker-entrypoint.…"   phpmyadmin   5 minutes ago   Up 4 minutes
      0.0.0.0:8091->80/tcp, [::]:8091->80/tcp
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
```

Retorna l'estat de cada servei en JSON:



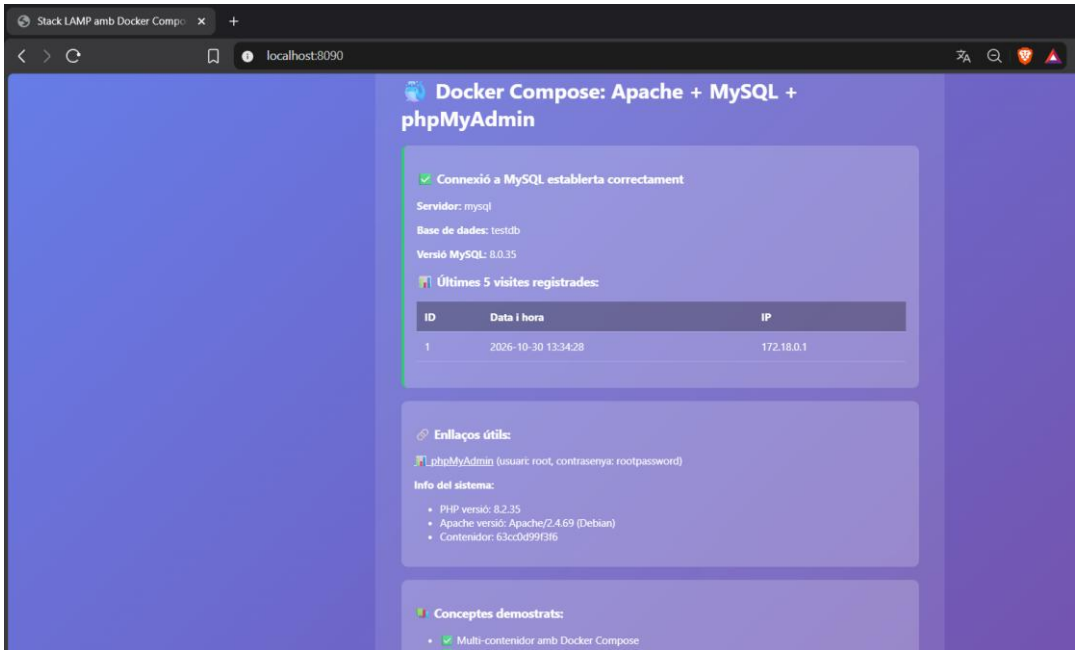
Amb docker compose exec apache bash entro dins del contenidor i comprovo que té PHP instal·lat. Provo d'escalar Apache a 3 contenidors, però no funciona perquè té un nom fix (container_name) i un port assignat. Per poder escalar, caldria eliminar aquestes dues configuracions.

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker compose exec apache bash
root@63cc0d99f3f6:/var/www/html# php -v
PHP 8.2.35 (cli) (built: Oct 28 2026 01:29:54) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.2.35, Copyright (c) Zend Technologies
    with Zend OPcache v8.2.35, Copyright (c), by Zend Technologies
root@63cc0d99f3f6:/var/www/html# ^C
root@63cc0d99f3f6:/var/www/html# quit
bash: quit: command not found
root@63cc0d99f3f6:/var/www/html# exit
exit

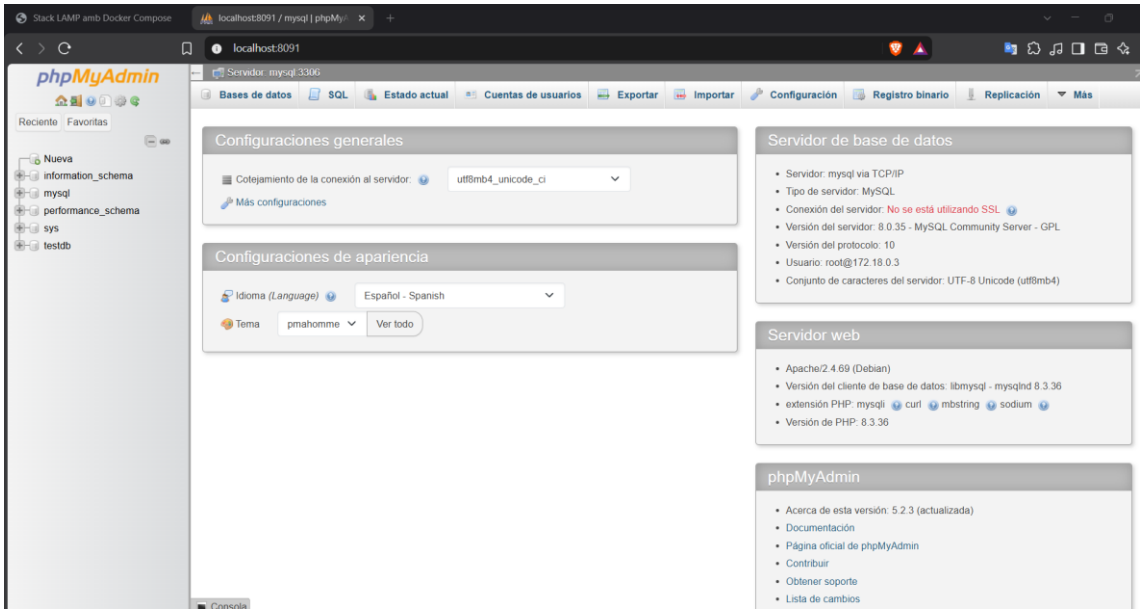
What's next:
    Debug this Compose error with Gordon → docker ai "help me fix this compose error"
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker compose up -d --scale apache=3
WARNING: The "apache" service is using the custom container name "lamp-apache". Docker requires each container to have a
 unique name. Remove the custom name to scale the service

What's next:
    Debug this Compose error with Gordon → docker ai "help me fix this compose error"
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
```

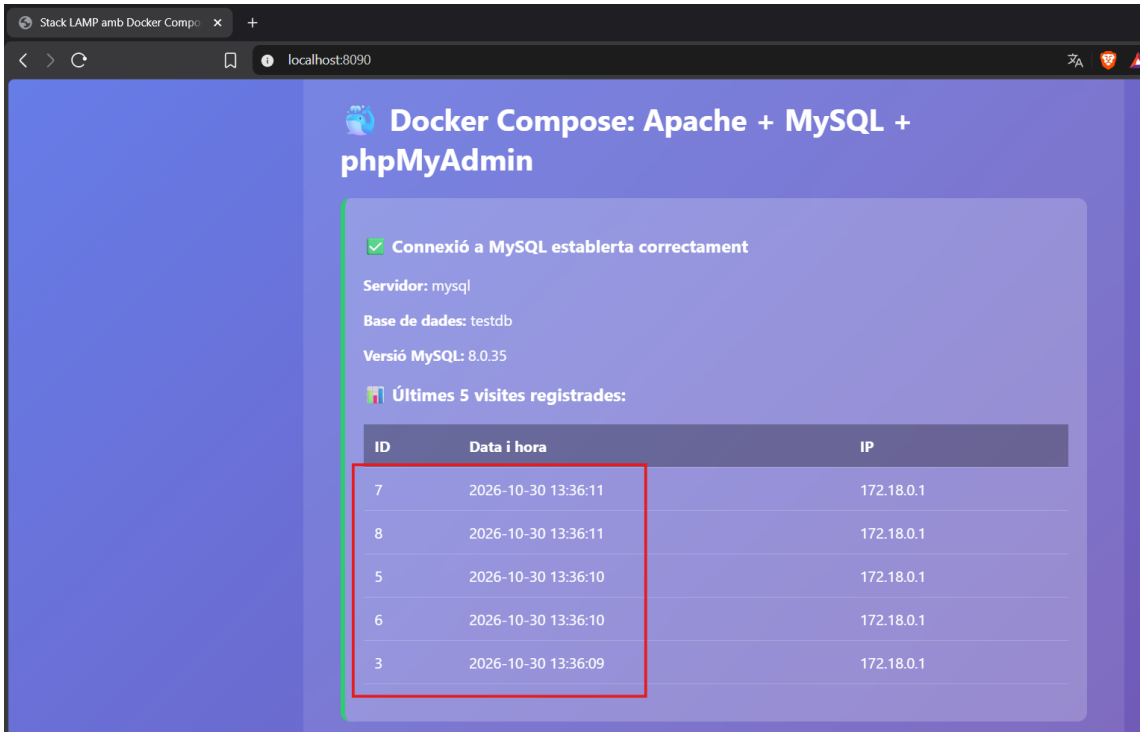
A http://localhost:8090 comprovo que la pàgina PHP es connecta correctament a MySQL. Es mostra la base de dades testdb, la versió 8.0.35 i la primera visita registrada amb la IP 172.18.0.1 (la gateway de la xarxa Docker).



A localhost:8091 (phpmyadmin) mostra el servidor mysql i les bases.



Després de recarregar localhost:8090 diverses vegades, la taula mostra les últimes 5 visites. Els ID no apareixen consecutius perquè es registren diverses visites en el mateix segon i es ordenen per timestamp, no per ID. Confirmació que les dades s'escriuen a MySql.



Xarxes i comunicació entre contenidors

El nom del servei es resol a la IP del contenidor Mysql. Apache PHP connecta amb Mysql per la xarxa Docker, llista information_schema, mysql...

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker compose exec apache getent hosts mysql
172.18.0.2      mysql
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker compose exec apache php -r "`$c = new mysqli('mysql', 'root', 'rootpassword', 'testdb'); echo `$c->server_info, PHP_EOL;"
8.0.35
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker compose exec mysql mysql -uroot -prootpassword -e "SHOW DATABASES;"
mysql: [Warning] Using a password on the command line interface can be insecure.
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| testdb             |
+--------------------+
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
```

Volums persistents vs Bind mounts

docker volume ls mostra el volum local, docker volume inspect... mostra un volum específic, docker volume prune elimina els volums (primer demana confirmació). Faig un backup, Docker descarrega alpine:latest i genera mysql-backup.tar.gz a la carpeta actual.

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker volume ls
DRIVER    VOLUME NAME
local     docker-compose-stack_mysql-data
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker volume inspect docker-compose-stack_mysql-data
[
    {
        "CreatedAt": "2026-10-30T12:21:01Z",
        "Driver": "local",
        "Labels": {
            "com.docker.compose.config-hash": "ebba52cf252a4c546fa26daff0064a1091d10f11049c0760a286aed36e82587d",
            "com.docker.compose.project": "docker-compose-stack",
            "com.docker.compose.version": "5.6.1",
            "com.docker.compose.volume": "mysql-data"
        },
        "Mountpoint": "/var/lib/docker/volumes/docker-compose-stack_mysql-data/_data",
        "Name": "docker-compose-stack_mysql-data",
        "Options": null,
        "Scope": "local"
    }
]
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker volume prune
WARNING! This will remove anonymous local volumes not used by at least one container.
Are you sure you want to continue? [y/N] n
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker run --rm -v docker-compose-stack_mysql-data:/data -v "${PWD}:/backup" alpine tar czf /backup
/mysql-backup.tar.gz -C /data .
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
7b229517ed22: Pull complete
3da1513d2124: Download complete
3ec36f3670df: Download complete
Digest: sha256:be2656061b536c1883c4034790e21d0a09c5aaaf51968b89368a3c0371ea2c9a
Status: Downloaded newer image for alpine:latest
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
```

Variables d'entorn i fitxers .env

L'estructura ara indica .env, docker-compose.yml i mysql-backup.tar.gz que és la còpia d'abans. El .env centralitza la configuració: credencials de Mysql.

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> tree /f
Listado de rutas de carpetas
El número de serie del volumen es 38C8-F55B
C:.
    .env
    docker-compose.yml
    mysql-backup.tar.gz

├───apache
    │   Dockerfile
    │
    ├───conf
    │       httpd.conf
    │
    └───html
            index.php

├───logs
        access.log
        error.log
        other_vhosts_access.log

└───mysql-data
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> cat .\.env
# ConfiguraciÃ³ de MySQL
MYSQL_ROOT_PASSWORD=supersecret2025
MYSQL_DATABASE=production_db
MYSQL_USER=webapp
MYSQL_PASSWORD=webapp_secret_2025

# Ports
APACHE_PORT=8090
PHPMYADMIN_PORT=8091
MYSQL_PORT=3306

# Timezone
TZ=Europe/Madrid
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
```

.gitignore conté .env perquè el fitxer amb credencials no es pugi al repositori.

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> cat .gitignore
.env
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
```

Modificació del docker-compose.yml per utilitzar variables:

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> cat .\docker-compose.yml
# DefiniciÃ³ dels serveis (contenidors)
services:

  # Servei Apache amb PHP
  apache:
    # Construeix la imatge des del Dockerfile
    build:
      context: ./apache
      dockerfile: Dockerfile

    # Nom del contenidor
    container_name: lamp-apache

    # Reinicia automÃ ticament si el contenidor falla
    restart: unless-stopped

    # Mapatge de ports: host:container
    ports:
      - "${APACHE_PORT}:80"

    # Volums: persistÃ¨ncia i desenvolupament
    volumes:
      - ./apache/html:/var/www/html  # Codi PHP (bind mount)
      - ./logs:/var/log/apache2      # Logs d'Apache

    # Variables d'entorn
    environment:
      - TZ=${TZ}

    # DepÃ¨n del servei MySQL: no arrenca fins que el healthcheck diu que estÃ  llest
    depends_on:
      mysql:
        condition: service_healthy

    # Xarxa per comunicar-se amb altres serveis
    networks:
      - lamp-network

    # LÃmits de recursos (opcional perÃ² recomanat)
    deploy:
      resources:
        limits:
          cpus: '1.0'
          memory: 512M

  # Servei MySQL
  mysql:
    # Imatge oficial de MySQL (versiÃ³ especÃfica, no "latest")
    image: mysql:8.0.35

    container_name: lamp-mysql
    restart: unless-stopped

    # Variables d'entorn per configurar MySQL
    environment:
      - MYSQL_ROOT_PASSWORD=${MYSQL_ROOT_PASSWORD}
      - MYSQL_DATABASE=${MYSQL_DATABASE}
      - MYSQL_USER=${MYSQL_USER}
      - MYSQL_PASSWORD=${MYSQL_PASSWORD}
      - TZ=${TZ}
```

```
      - MYSQL_PASSWORD=${MYSQL_PASSWORD}
      - TZ=${TZ}

    # Volum amb nom per persistir dades de MySQL
    volumes:
      - mysql-data:/var/lib/mysql

    # Exposa el port MySQL (opcional, nomÃ©s si vols accedir des de l'host)
    ports:
      - "${MYSQL_PORT}:3306"

    networks:
      - lamp-network

    # Healthcheck per verificar que MySQL estÃ  llest
    healthcheck:
      test: ["CMD", "mysqladmin", "ping", "-h", "localhost", "-uroot", "-p${MYSQL_ROOT_PASSWORD}"]
      interval: 10s
      timeout: 5s
      retries: 3
      start_period: 30s

  # Servei phpMyAdmin (interfÃ-cie web per gestionar MySQL)
  phpmyadmin:
    image: phpmyadmin:5.2-apache
    container_name: lamp-phpmyadmin
    restart: unless-stopped

    ports:
      - "${PHPMYADMIN_PORT}:80"   # el 8081 ja l'ocupa el contenidor apache-dockerfile de la Part 2

    environment:
      - PMA_HOST=mysql
      - PMA_PORT=3306
      - PMA_USER=root
      - PMA_PASSWORD=${MYSQL_ROOT_PASSWORD}
      - UPLOAD_LIMIT=50M

    depends_on:
      - mysql

    networks:
      - lamp-network

# DefiniciÃ³ de xarxes
networks:
  lamp-network:
    driver: bridge  # Xarxa bridge per defecte (permet comunicaciÃ³ entre contenidors)

# DefiniciÃ³ de volums amb nom (persistents)
volumes:
  mysql-data:
    driver: local  # Volum local gestionat per Docker
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
```

Docker compose config OK

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker compose config
name: docker-compose-stack
services:
  apache:
    deploy:
      resources:
        limits:
          cpus: 1
          memory: "536870912"
    restart: unless-stopped
    container_name: lamp-apache
    environment:
      TZ: Europe/Madrid
    networks:
      lamp-network: null
    volumes:
      - type: bind
        source: C:\Users\USER\docker-apache-lab\docker-compose-stack\apache\html
        target: /var/www/html
        bind: {}
      - type: bind
        source: C:\Users\USER\docker-apache-lab\docker-compose-stack\logs
        target: /var/log/apache2
        bind: {}
    build:
      context: C:\Users\USER\docker-apache-lab\docker-compose-stack\apache
```

Repte 5

Afegir servei Redis al docker-compose.yml

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> cat .\docker-compose.yml
# DefiniciÃ³ dels serveis (contenidors)
services:

  redis:
    image: redis:7-alpine
    container_name: lamp-redis
    restart: unless-stopped
    ports:
      - "6379:6379"
    volumes:
      - redis-data:/data
    command: redis-server --appendonly yes
    networks:
      - lamp-network

  # Servei Apache amb PHP
  apache:
    # Construeix la imatge des del Dockerfile
    build:
```

A la secció volumes afegeixo redis-data amb driver: local, al costat de mysql-data.

```
# DefiniciÃ³ de volums amb nom (persistents)
volumes:
  mysql-data:
    driver: local  # Volum local gestionat per Docker
  redis-data:
    driver: local
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
```

Instal·lo la llibreria Redis per a PHP amb pecl install redis i l'activo juntament amb mysqli.

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> cat .\apache\Dockerfile
FROM php:8.2-apache

LABEL maintainer="s.martinez6@sapalomera.cat"
LABEL description="Apache amb PHP, MySQL i Redis"

# InstalÂ·la les extensions mysqli i redis
RUN docker-php-ext-install mysqli && pecl install redis && docker-php-ext-enable mysqli redis

# Activa mod_rewrite
RUN a2enmod rewrite

# Copia el contingut web
COPY html/ /var/www/html/

# Exposa el port 80
EXPOSE 80
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
```

cat .\apache\html\index.php: Mostra el nom, comptador de visites amb Redis i nombre de visites, si falla mostra un missatge d'error. La connexió amb Misql continua igual.

```
<body>
    <div class="container">
        <h1>ðŸ³ Docker Compose: Apache + MySQL + phpMyAdmin</h1>

        <?php
        // connexiÃ³ Redis
        try {
            $redis = new Redis();
            $redis->connect('redis', 6379, 2);

            // Incrementa el comptador de visites
            $visites = $redis->incr('visites:shaila');

            echo '<div class="status success">';
            echo '<h2>Shaila MartÃnez</h2>';
            echo '<h3>Comptador de visites amb Redis</h3>';
            echo '<p><strong>Nombre de visites:</strong> ' . $visites . '</p>';
            echo '</div>';

            $redis->close();

        } catch (RedisException $e) {
            echo '<div class="status error">';
            echo '<h3>Error de connexiÃ³ a Redis</h3>';
            echo '<p>' . htmlspecialchars($e->getMessage()) . '</p>';
            echo '</div>';
        }



        // ConfiguraciÃ³ de la connexiÃ³ a MySQL
        $host = 'mysql';  // Nom del servei en docker-compose.yml
        $user = 'root';
```

cat .\docker-compose.yml: He afegit aquesta condició perquè Apache arrenqui després de Redis.

```
    # Variables d'entorn
    environment:
      - TZ=${TZ}

    # DepÃ¨n del servei MySQL: no arrenca fins que el healthcheck diu que estÃ  llest
    depends_on:
      mysql:
        condition: service_healthy
      redis:
        condition: service_started

    # Xarxa per comunicar-se amb altres serveis
    networks:
      - lamp-network

    # LÃmits de recursos (opcional perÃ² recomanat)
    deploy:
      resources:
        limits:
          cpus: '1.0'
          memory: 512M
```

Validació del fitxer:

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker compose config
name: docker-compose-stack
services:
  apache:
    deploy:
      resources:
        limits:
          cpus: 1
          memory: "536870912"
    restart: unless-stopped
    container_name: lamp-apache
    environment:
      TZ: Europe/Madrid
    networks:
      lamp-network: null
    volumes:
      - type: bind
        source: C:\Users\USER\docker-apache-lab\docker-compose-stack\apache\html
        target: /var/www/html
        bind: {}
      - type: bind
        source: C:\Users\USER\docker-apache-lab\docker-compose-stack\logs
        target: /var/log/apache2
        bind: {}
    build:
      context: C:\Users\USER\docker-apache-lab\docker-compose-stack\apache
```

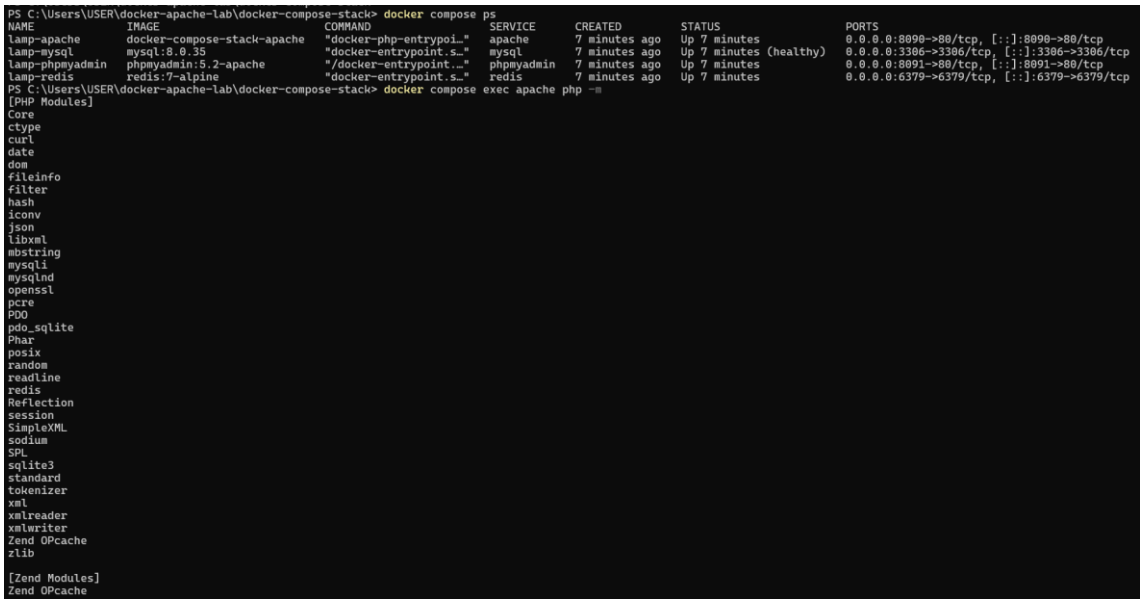
Comprovació de que ara hi ha els 4 serveis, entre ells Redis.

```
Windows PowerShell
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker compose config --services
mysql
phpmyadmin
redis
apache
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
```

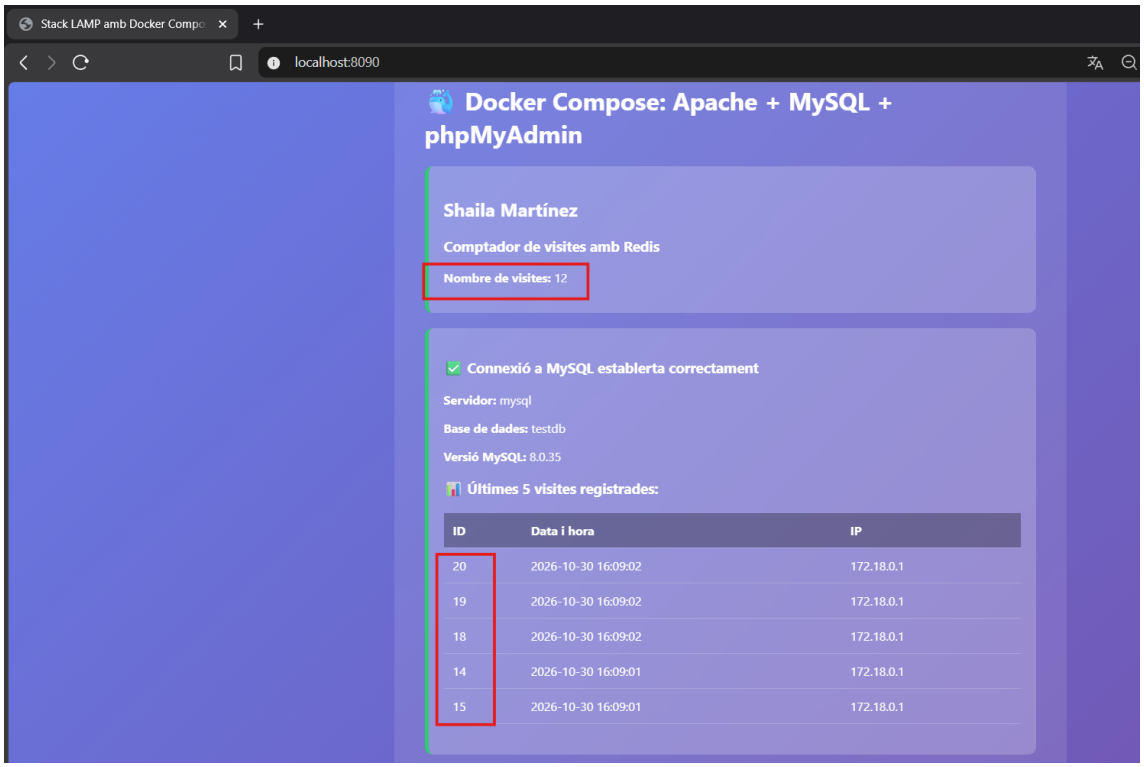
Docker descarrega redis:7-alpine, reconstrueix la imatge d'Apache. Arrenca els 4 contenidors.

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker compose up -d --build
[+] up 10/10
 ✔Image redis:7-alpine Pulled                                                                                    3.0s
[+] Building 26.6s (12/12) FINISHED
 => [internal] load local bake definitions                                                                       0.0s
 => => reading from stdin 607B                                                                                   0.0s
 => [internal] load build definition from Dockerfile                                                             0.0s
 => => transferring dockerfile: 433B                                                                             0.0s
 => [internal] load metadata for docker.io/library/php:8.2-apache                                                0.9s
 => [auth] library/php:pull token for registry-1.docker.io                                                       0.0s
 => [internal] load .dockerignore                                                                                0.0s
 => => transferring context: 2B                                                                                  0.0s
 => CACHED [1/4] FROM docker.io/library/php:8.2-apache@sha256:7b7574ed556f3cd450aafbdcb1424acdfbd67d540ce50c2c9f0  0.1s
 => => resolve docker.io/library/php:8.2-apache@sha256:7b7574ed556f3cd450aafbdcb1424acdfbd67d540ce50c2c9f03e5698b  0.0s
 => [internal] load build context                                                                                0.0s
 => => transferring context: 5.89kB                                                                              0.0s
 => [2/4] RUN docker-php-ext-install mysqli && pecl install redis && docker-php-ext-enable mysqli redis         24.3s
 => [3/4] RUN a2enmod rewrite                                                                                    0.3s
 => [4/4] COPY html/ /var/www/html/                                                                              0.1s
 => exporting to image                                                                                           0.6s
 => => exporting layers                                                                                          0.3s
 => => exporting manifest sha256:21bcb4190dc63e4de6d0533643e7133cc29657d82625533ce4751215140a1c7d                0.0s
 => => exporting config sha256:e49191b1ac1d864a06fc83116ad37ad21e6b72f54612c9685b6d1d70beaf35eb                  0.0s
 => => exporting attestation manifest sha256:ba88a17235bcd0ca91a1c699786a7ac00816b4a2548f4767ae628fe7d024a033    0.0s
 => => exporting manifest list sha256:0f698c1045dc9271bc76b48bc4a06f5cf07a10e125cb065be93b6cf2a057eccd           0.0s
 => => naming to docker.io/library/docker-compose-stack-apache:latest                                            0.0s
 => => unpacking to docker.io/library/docker-compose-stack-apache:latest                                         0.1s
 => resolving provenance for metadata file                                                                       0.0s
[+] up 16/16
 ✔Image redis:7-alpine                  Pulled                                                                   3.0s
 ✔Image docker-compose-stack-apache     Built                                                                   26.9s
 ✔Volume docker-compose-stack_redis-data Created                                                                 0.0s
 ✔Container lamp-mysql                  Healthy                                                                  8.5s
 ✔Container lamp-redis                  Started                                                                  3.5s
 ✔Container lamp-phpmyadmin             Started                                                                  2.5s
 ✔Container lamp-apache                 Started                                                                  7.5s
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
```

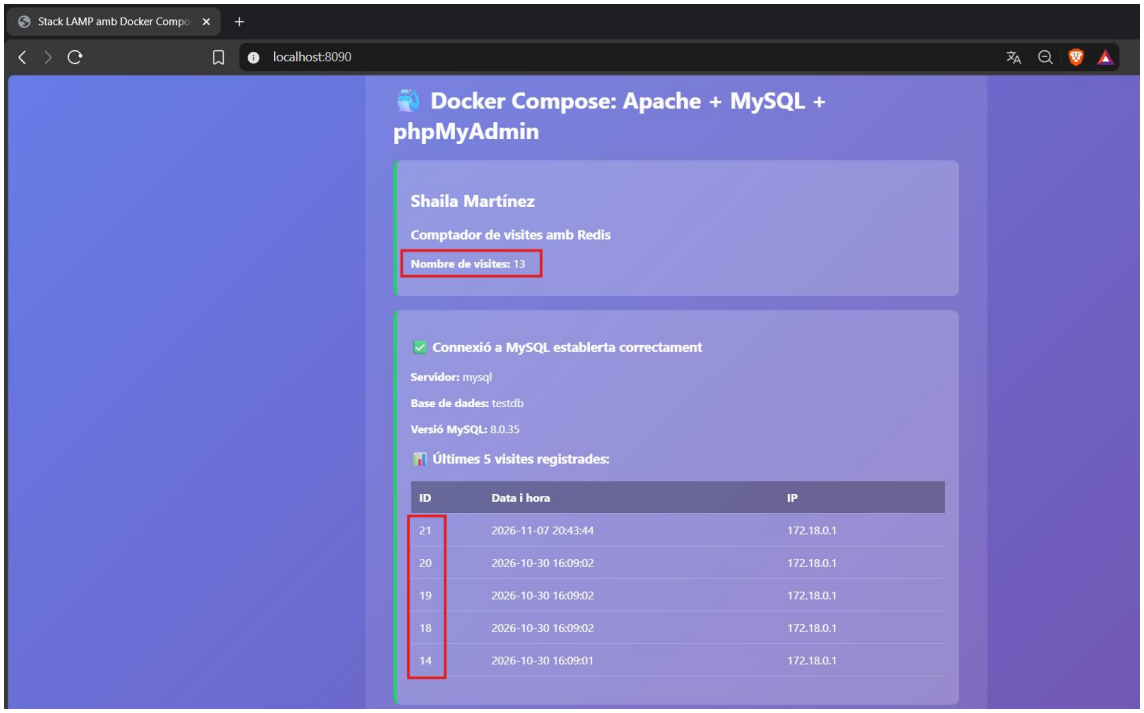
Els 4 serveis en marxa i mòduls de PHP, extensió correctament instal·lada.



Localhost:8090 surt el nombre de visites i a sota la connexió a MySQL amb les últimes cinc visites.



Cada vegada que actualitzo la pàgina, el comptador de Redis augmenta (de 12 a 13). Això comprova que Redis funciona correctament i que Apache s'hi connecta mitjançant el nom del servei.



Volums:

```
PS C:\Users\USER\docker-apache-lab\docker-compose-stack> docker volume ls
DRIVER    VOLUME NAME
local     docker-compose-stack_mysql-data
local     docker-compose-stack_redis-data
PS C:\Users\USER\docker-apache-lab\docker-compose-stack>
```